

Supplementary Technical Report: Revised Leakage-Free Implementation of ClinicalEarlyCancerDF

Supplementary Note S1.

S1.1 Leakage-Free Nested Evaluation Protocol

To address the methodological concern regarding potential information leakage during model development, the ClinicalEarlyCancerDF framework was completely re-implemented using a leakage-free nested cross-validation protocol, as illustrated in **Supplementary Figure S1**.

Unlike the original implementation, the revised workflow first partitions the complete CancerSEEK dataset into **outer training** and **outer test** partitions using stratified outer cross-validation. The unseen outer test fold is strictly isolated and is never used during any stage of feature selection, class balancing, hyper-parameter optimization, or learner selection.

All adaptive learning components are fitted exclusively within the outer training partition. Consequently, each outer test fold serves as an independent dataset for unbiased estimation of the framework's generalization performance.

S1.2 Adaptive Computational Pipeline

As shown in **Supplementary Figure S1**, the proposed ClinicalEarlyCancerDF framework consists of four adaptive computational modules.

Step 1: Data Preprocessing

The outer training partition first undergoes data preprocessing, including:

- missing-value imputation,
- one-hot encoding of categorical variables, and
- data normalization when required.

These preprocessing operations are fitted exclusively on the training partition and subsequently applied to the corresponding outer test fold, preventing any information leakage.

Step 2: Best Recursive Feature Elimination (bRFE)

The preprocessed training data are then analyzed using the proposed **Best Recursive Feature Elimination (bRFE)** algorithm.

Unlike conventional RFE approaches that employ a single learning algorithm, bRFE evaluates two candidate feature-selection learners:

- XGBoost-RFE
- Random Forest-RFE

For each learner, recursive feature elimination iteratively removes the least important feature until only two features remain. At every iteration, the remaining feature subset is evaluated using inner cross-validation. The feature subset producing the highest validation ROC-AUC is recorded as the optimal subset for that learner.

After both learners complete the recursive elimination process, bRFE compares their best-performing subsets and returns the feature subset that achieves the highest validation performance. Consequently, different outer training partitions may produce different optimal biomarker subsets depending on the underlying data characteristics.

Step 3: Efficient SMOTE-based Oversampling Method (ESOM)

The selected biomarker subset is subsequently analyzed by the proposed Efficient SMOTE-based Oversampling Method (ESOM).

Unlike conventional oversampling methods that are always applied, ESOM first estimates the class imbalance ratio within the outer training partition.

- If the imbalance ratio is less than or equal to 5%, no oversampling is performed.
- Otherwise, Adaptive SMOTE is applied to generate only the required number of minority samples until class balance is achieved.

This strategy avoids unnecessary oversampling while preserving the original data distribution whenever possible.

Step 4: Ensemble Optimization Learning Framework (EOLF)

The balanced training data are finally processed by the proposed Ensemble Optimization Learning Framework (EOLF).

Three candidate prediction algorithms are considered:

- Random Forest
- XGBoost
- LightGBM

Each learner undergoes independent hyper-parameter optimization using GridSearchCV within the inner cross-validation loop. The learner achieving the highest validation ROC-AUC is selected as the final prediction model for the corresponding outer fold.

The optimized classifier is then retrained using the complete outer training partition and evaluated only once on the previously unseen outer test fold.

The entire procedure is repeated across all outer folds to obtain an unbiased estimate of model generalization.

S1.3 Experimental Results

The revised leakage-free implementation was evaluated using **10-fold nested cross-validation** on the CancerSEEK dataset.

The average predictive performance across all outer folds was:

Metric	Mean $\pm$ SD
Accuracy	0.983 $\pm$ 0.007
ROC-AUC	0.998 $\pm$ 0.003
PR-AUC	0.998 $\pm$ 0.003
Precision	0.983 $\pm$ 0.013
Recall	0.979 $\pm$ 0.010
Specificity	0.986 $\pm$ 0.011
F1-score	0.981 $\pm$ 0.007

The small standard deviations across all performance metrics indicate that the framework maintained highly consistent predictive performance on unseen data despite repeated changes in the training partitions.

S1.4 Observations from the Nested Evaluation

The revised implementation revealed several important observations that further support the adaptive design of ClinicalEarlyCancerDF.

Observation 1. Adaptive Feature Selection

Across the ten outer folds, the proposed bRFE algorithm selected different optimal biomarker subsets ranging from **15 to 24 features**.

This variation demonstrates that the framework does not rely on a fixed biomarker panel. Instead, the selected feature subset is automatically determined according to the statistical characteristics of each training partition.

Observation 2. Adaptive Learner Selection

Similarly, EOLF selected different prediction models across the outer folds.

LightGBM was selected in most folds, whereas XGBoost was selected in folds where it achieved superior validation performance.

This behavior demonstrates that ClinicalEarlyCancerDF is not constrained to a predetermined classifier but instead adapts its prediction model according to the characteristics of the training data.

Observation 3. Stable Generalization

Although both the selected feature subsets and prediction models varied across outer folds, the predictive performance remained remarkably stable.

The consistently high ROC-AUC together with the small fold-to-fold standard deviations indicate that the proposed adaptive strategy captures robust biological signals rather than overfitting to a specific feature subset or learning algorithm.

Observation 4. Framework Adaptability

The fold-wise variation in selected biomarkers and prediction models provides direct experimental evidence that ClinicalEarlyCancerDF behaves as an adaptive computational intelligence framework.

Rather than learning a single static prediction model, the framework automatically constructs an optimized computational pipeline for each training partition through adaptive feature selection, conditional imbalance handling, and adaptive learner optimization.

This adaptive behavior represents one of the principal methodological contributions of the proposed framework.

Overall Findings

The revised implementation confirms that ClinicalEarlyCancerDF maintains excellent predictive performance under a rigorous leakage-free nested cross-validation protocol. More importantly, the experiments demonstrate that the framework adaptively selects feature subsets, class balancing strategies, and prediction models according to the characteristics of each training partition while consistently achieving stable performance on previously unseen data. The observed variation across outer folds reflects the adaptive design of the framework rather than instability, providing additional evidence that ClinicalEarlyCancerDF functions as an adaptive computational intelligence framework for liquid biopsy-based early cancer detection.

Fold	bRFE Learner	Features	ESOM	Final Learner	Accuracy	ROC-AUC	PR-AUC
1	XGB	17	Yes	LightGBM	0.978	0.999	0.999
2	XGB	15	Yes	LightGBM	0.973	0.991	0.989
3	XGB	19	Yes	LightGBM	0.989	1.000	1.000
4	XGB	20	Yes	XGBoost	0.989	0.999	0.999
5	XGB	16	Yes	LightGBM	0.978	0.997	0.995
6	XGB	19	Yes	LightGBM	0.984	1.000	1.000
7	XGB	17	Yes	LightGBM	0.984	1.000	0.999
8	XGB	20	Yes	LightGBM	0.978	0.999	0.999
9	XGB	24	Yes	XGBoost	0.983	1.000	1.000
10	XGB	19	Yes	LightGBM	0.994	1.000	1.000

Supplementary Table S2

Overall Nested Cross-Validation Performance

Metric	Mean $\pm$ SD
Accuracy	0.983 $\pm$ 0.007
ROC-AUC	0.998 $\pm$ 0.003
PR-AUC	0.998 $\pm$ 0.003
Precision	0.983 $\pm$ 0.013
Recall	0.979 $\pm$ 0.010
Specificity	0.986 $\pm$ 0.011
F1-score	0.981 $\pm$ 0.007

These results were obtained under a fully leakage-free nested cross-validation protocol in which feature selection, class balancing, hyperparameter optimization, and learner selection were performed exclusively within the outer training partition. Consequently, the reported performance represents an unbiased estimate of the framework's generalization capability.

Supplementary Table S3

Fold-wise Prediction Performance

Fold	Accuracy	ROC-AUC	Precision	Recall	Specificity	F1-score	PR-AUC
1	0.978	0.999	0.975	0.975	0.980	0.975	0.999
2	0.973	0.991	0.952	0.988	0.960	0.970	0.989
3	0.989	1.000	1.000	0.975	1.000	0.988	1.000
4	0.989	0.999	0.988	0.988	0.990	0.988	0.999
5	0.978	0.997	0.975	0.975	0.980	0.975	0.995
6	0.984	1.000	0.988	0.975	0.990	0.981	1.000
7	0.984	1.000	0.988	0.976	0.990	0.982	0.999
8	0.978	0.999	0.988	0.963	0.990	0.975	0.999
9	0.983	1.000	0.988	0.975	0.990	0.981	1.000
10	0.994	1.000	0.988	1.000	0.990	0.994	1.000

Supplementary Table S4

Adaptive Feature Selection Summary

Fold	Selected Features
1	17
2	15
3	19
4	20
5	16
6	19
7	17
8	20
9	24
10	19

Summary

Statistic	Value
Minimum	15
Maximum	24
Average	18.6
Standard Deviation	2.55

Observation

The optimal biomarker subset varied across outer folds, demonstrating that bRFE adaptively selected dataset-specific biomarkers according to the characteristics of each training partition.

Supplementary Table S5

Adaptive Learner Selection

Fold	Final Learner
1	LightGBM
2	LightGBM
3	LightGBM
4	XGBoost
5	LightGBM
6	LightGBM
7	LightGBM
8	LightGBM
9	XGBoost
10	LightGBM

Learner Selection Frequency

Learner	Selected Folds	Frequency
LightGBM	1,2,3,5,6,7,8,10	8
XGBoost	4,9	2
Random Forest	—	0

Observation

EOLF dynamically selected the best-performing learning algorithm based on the characteristics of each training partition. The selection of different learners across folds further demonstrates the adaptive nature of the proposed framework.

Supplementary Table S6

GridSearchCV Optimization Summary

Fold	Selected Learner	Hyperparameter Search	Selected Hyperparameters
1	LightGBM	GridSearchCV	See Supplementary File S1
2	LightGBM	GridSearchCV	See Supplementary File S1
3	LightGBM	GridSearchCV	See Supplementary File S1
4	XGBoost	GridSearchCV	See Supplementary File S1
5	LightGBM	GridSearchCV	See Supplementary File S1
6	LightGBM	GridSearchCV	See Supplementary File S1
7	LightGBM	GridSearchCV	See Supplementary File S1
8	LightGBM	GridSearchCV	See Supplementary File S1
9	XGBoost	GridSearchCV	See Supplementary File S1
10	LightGBM	GridSearchCV	See Supplementary File S1

Supplementary File S1

Complete Leakage-Free Nested Cross-Validation Execution Log

Includes:

- Fold-by-fold bRFE execution.
- Recursive feature elimination process.
- ESOM class imbalance analysis.
- GridSearchCV optimization.
- EOLF learner selection.
- Final evaluation metrics.
- Overall performance summary.

Supplementary Notebook S1

ClinicalEarlyCancerDF: Complete Reproducible Implementation

Contents:

1. Data loading.
2. Data preprocessing.
3. Leakage-free nested cross-validation.
4. bRFE implementation.
5. ESOM implementation.
6. EOLF implementation.
7. GridSearchCV optimization.
8. Final evaluation.
9. Statistical summary.
10. Visualization.
11. Result export.

Supplementary Run Logs S1

Leakage-Free Nested Cross-Validation Evaluation of ClinicalEarlyCancerDF Run logs

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ClinicalEarlyCancerDF
Nested Cross Validation

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Outer Fold 1

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Running bRFE...

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Best Recursive Feature Elimination (bRFE)

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Evaluating learner: XGB

Remaining Features: 53 | AUC=0.9978 | ACC=0.9872
Remaining Features: 52 | AUC=0.9977 | ACC=0.9829
Remaining Features: 51 | AUC=0.9979 | ACC=0.9872
Remaining Features: 50 | AUC=0.9980 | ACC=0.9853
Remaining Features: 49 | AUC=0.9981 | ACC=0.9872
Remaining Features: 48 | AUC=0.9978 | ACC=0.9841
Remaining Features: 47 | AUC=0.9978 | ACC=0.9884
Remaining Features: 46 | AUC=0.9980 | ACC=0.9841
Remaining Features: 45 | AUC=0.9979 | ACC=0.9810
Remaining Features: 44 | AUC=0.9982 | ACC=0.9847
Remaining Features: 43 | AUC=0.9979 | ACC=0.9866
Remaining Features: 42 | AUC=0.9981 | ACC=0.9847
Remaining Features: 41 | AUC=0.9976 | ACC=0.9866
Remaining Features: 40 | AUC=0.9982 | ACC=0.9878
Remaining Features: 39 | AUC=0.9978 | ACC=0.9841
Remaining Features: 38 | AUC=0.9979 | ACC=0.9872
Remaining Features: 37 | AUC=0.9976 | ACC=0.9853

Remaining Features: 36	AUC=0.9977	ACC=0.9866
Remaining Features: 35	AUC=0.9978	ACC=0.9853
Remaining Features: 34	AUC=0.9978	ACC=0.9853
Remaining Features: 33	AUC=0.9981	ACC=0.9835
Remaining Features: 32	AUC=0.9976	ACC=0.9859
Remaining Features: 31	AUC=0.9980	ACC=0.9859
Remaining Features: 30	AUC=0.9980	ACC=0.9841
Remaining Features: 29	AUC=0.9977	ACC=0.9841
Remaining Features: 28	AUC=0.9978	ACC=0.9872
Remaining Features: 27	AUC=0.9980	ACC=0.9859
Remaining Features: 26	AUC=0.9981	ACC=0.9847
Remaining Features: 25	AUC=0.9980	ACC=0.9847
Remaining Features: 24	AUC=0.9982	ACC=0.9841
Remaining Features: 23	AUC=0.9982	ACC=0.9841
Remaining Features: 22	AUC=0.9980	ACC=0.9823
Remaining Features: 21	AUC=0.9982	ACC=0.9866
Remaining Features: 20	AUC=0.9978	ACC=0.9847
Remaining Features: 19	AUC=0.9979	ACC=0.9829
Remaining Features: 18	AUC=0.9983	ACC=0.9847
Remaining Features: 17	AUC=0.9982	ACC=0.9884
Remaining Features: 16	AUC=0.9980	ACC=0.9884
Remaining Features: 15	AUC=0.9978	ACC=0.9866
Remaining Features: 14	AUC=0.9981	ACC=0.9841
Remaining Features: 13	AUC=0.9975	ACC=0.9847
Remaining Features: 12	AUC=0.9972	ACC=0.9804
Remaining Features: 11	AUC=0.9967	ACC=0.9755
Remaining Features: 10	AUC=0.9969	ACC=0.9743
Remaining Features: 9	AUC=0.9969	ACC=0.9762
Remaining Features: 8	AUC=0.9971	ACC=0.9798
Remaining Features: 7	AUC=0.9958	ACC=0.9737
Remaining Features: 6	AUC=0.9959	ACC=0.9749
Remaining Features: 5	AUC=0.9960	ACC=0.9749
Remaining Features: 4	AUC=0.9847	ACC=0.9523
Remaining Features: 3	AUC=0.9597	ACC=0.9083
Remaining Features: 2	AUC=0.8171	ACC=0.7861

XGB Best Subset

Features : 17
Best AUC : 0.998
Best ACC : 0.9884

Evaluating learner: RF

Remaining Features: 53	AUC=0.9966	ACC=0.9755
Remaining Features: 52	AUC=0.9967	ACC=0.9749
Remaining Features: 51	AUC=0.9965	ACC=0.9755
Remaining Features: 50	AUC=0.9963	ACC=0.9737
Remaining Features: 49	AUC=0.9968	ACC=0.9719
Remaining Features: 48	AUC=0.9963	ACC=0.9725
Remaining Features: 47	AUC=0.9965	ACC=0.9713
Remaining Features: 46	AUC=0.9964	ACC=0.9737
Remaining Features: 45	AUC=0.9964	ACC=0.9713
Remaining Features: 44	AUC=0.9969	ACC=0.9694
Remaining Features: 43	AUC=0.9968	ACC=0.9737
Remaining Features: 42	AUC=0.9961	ACC=0.9707
Remaining Features: 41	AUC=0.9964	ACC=0.9731
Remaining Features: 40	AUC=0.9963	ACC=0.9749
Remaining Features: 39	AUC=0.9967	ACC=0.9743
Remaining Features: 38	AUC=0.9969	ACC=0.9737
Remaining Features: 37	AUC=0.9968	ACC=0.9719
Remaining Features: 36	AUC=0.9967	ACC=0.9707
Remaining Features: 35	AUC=0.9966	ACC=0.9713
Remaining Features: 34	AUC=0.9967	ACC=0.9725
Remaining Features: 33	AUC=0.9973	ACC=0.9762
Remaining Features: 32	AUC=0.9967	ACC=0.9713

Remaining Features: 31 | AUC=0.9972 | ACC=0.9762
Remaining Features: 30 | AUC=0.9973 | ACC=0.9755
Remaining Features: 29 | AUC=0.9968 | ACC=0.9743
Remaining Features: 28 | AUC=0.9966 | ACC=0.9743
Remaining Features: 27 | AUC=0.9966 | ACC=0.9749
Remaining Features: 26 | AUC=0.9958 | ACC=0.9676
Remaining Features: 25 | AUC=0.9960 | ACC=0.9713
Remaining Features: 24 | AUC=0.9957 | ACC=0.9694
Remaining Features: 23 | AUC=0.9962 | ACC=0.9694
Remaining Features: 22 | AUC=0.9963 | ACC=0.9731
Remaining Features: 21 | AUC=0.9963 | ACC=0.9719
Remaining Features: 20 | AUC=0.9956 | ACC=0.9731
Remaining Features: 19 | AUC=0.9953 | ACC=0.9713
Remaining Features: 18 | AUC=0.9962 | ACC=0.9737
Remaining Features: 17 | AUC=0.9958 | ACC=0.9713
Remaining Features: 16 | AUC=0.9958 | ACC=0.9707
Remaining Features: 15 | AUC=0.9956 | ACC=0.9670
Remaining Features: 14 | AUC=0.9954 | ACC=0.9700
Remaining Features: 13 | AUC=0.9950 | ACC=0.9682
Remaining Features: 12 | AUC=0.9943 | ACC=0.9639
Remaining Features: 11 | AUC=0.9937 | ACC=0.9652
Remaining Features: 10 | AUC=0.9935 | ACC=0.9621
Remaining Features: 9 | AUC=0.9930 | ACC=0.9597
Remaining Features: 8 | AUC=0.9900 | ACC=0.9499
Remaining Features: 7 | AUC=0.9843 | ACC=0.9425
Remaining Features: 6 | AUC=0.9750 | ACC=0.9181
Remaining Features: 5 | AUC=0.9763 | ACC=0.9156
Remaining Features: 4 | AUC=0.9747 | ACC=0.9120
Remaining Features: 3 | AUC=0.9720 | ACC=0.9150
Remaining Features: 2 | AUC=0.9641 | ACC=0.9089

RF Best Subset

Features : 28
Best AUC : 0.9966
Best ACC : 0.9749

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bRFE Completed

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Selected Learner : XGB
Selected Features: 17
Best AUC : 0.9980
Best Accuracy : 0.9884

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Running ESOM...

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Efficient SMOTE-based Oversampling Method (ESOM)

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Original Class Distribution
Class 0: 905
Class 1: 731

Majority Class : 0
Minority Class : 1
Class Imbalance: 19.23%

Applying Adaptive SMOTE
k_neighbors = 5

Balanced Class Distribution

Class 0: 905

Class 1: 905

Generated Samples : 174

Running EOLF...

Evaluating : RF

Best AUC : 0.9967

Best ACC : 0.979

Evaluating : XGB

Best AUC : 0.9984

Best ACC : 0.9834

Evaluating : LightGBM

Best AUC : 0.9989

Best ACC : 0.9878

EOLF Completed

	Model	AUC	Accuracy \
0	RF	0.996716	0.979006
1	XGB	0.998364	0.983425
2	LightGBM	0.998871	0.987845

BestParameters

0	{'max_depth': None, 'min_samples_leaf': 1, 'mi...
1	{'learning_rate': 0.1, 'max_depth': 3, 'n_esti...
2	{'learning_rate': 0.1, 'n_estimators': 200, 'n...

Selected Model : LightGBM

Best AUC : 0.9989

Best ACC : 0.9878

Training Final Model...

Fold Results

Model : LightGBM

Features : 17

Accuracy : 0.9780

AUC : 0.9994

Precision : 0.9753

Recall : 0.9753

Specificity : 0.9802

F1-score : 0.9753

PR-AUC : 0.9992

Outer Fold 2

Running bRFE...

Best Recursive Feature Elimination (bRFE)

Evaluating learner: XGB

Remaining Features: 53 | AUC=0.9978 | ACC=0.9847

Remaining Features: 52 | AUC=0.9979 | ACC=0.9841
Remaining Features: 51 | AUC=0.9980 | ACC=0.9841
Remaining Features: 50 | AUC=0.9983 | ACC=0.9829
Remaining Features: 49 | AUC=0.9979 | ACC=0.9847
Remaining Features: 48 | AUC=0.9980 | ACC=0.9841
Remaining Features: 47 | AUC=0.9978 | ACC=0.9841
Remaining Features: 46 | AUC=0.9983 | ACC=0.9841
Remaining Features: 45 | AUC=0.9983 | ACC=0.9835
Remaining Features: 44 | AUC=0.9982 | ACC=0.9853
Remaining Features: 43 | AUC=0.9981 | ACC=0.9847
Remaining Features: 42 | AUC=0.9981 | ACC=0.9829
Remaining Features: 41 | AUC=0.9979 | ACC=0.9817
Remaining Features: 40 | AUC=0.9984 | ACC=0.9835
Remaining Features: 39 | AUC=0.9981 | ACC=0.9841
Remaining Features: 38 | AUC=0.9979 | ACC=0.9866
Remaining Features: 37 | AUC=0.9981 | ACC=0.9829
Remaining Features: 36 | AUC=0.9980 | ACC=0.9847
Remaining Features: 35 | AUC=0.9984 | ACC=0.9847
Remaining Features: 34 | AUC=0.9981 | ACC=0.9841
Remaining Features: 33 | AUC=0.9980 | ACC=0.9841
Remaining Features: 32 | AUC=0.9981 | ACC=0.9835
Remaining Features: 31 | AUC=0.9983 | ACC=0.9811
Remaining Features: 30 | AUC=0.9982 | ACC=0.9823
Remaining Features: 29 | AUC=0.9981 | ACC=0.9847
Remaining Features: 28 | AUC=0.9982 | ACC=0.9847
Remaining Features: 27 | AUC=0.9982 | ACC=0.9829
Remaining Features: 26 | AUC=0.9980 | ACC=0.9841
Remaining Features: 25 | AUC=0.9981 | ACC=0.9835
Remaining Features: 24 | AUC=0.9981 | ACC=0.9841
Remaining Features: 23 | AUC=0.9982 | ACC=0.9872
Remaining Features: 22 | AUC=0.9981 | ACC=0.9872
Remaining Features: 21 | AUC=0.9982 | ACC=0.9847
Remaining Features: 20 | AUC=0.9980 | ACC=0.9841
Remaining Features: 19 | AUC=0.9979 | ACC=0.9859
Remaining Features: 18 | AUC=0.9977 | ACC=0.9853
Remaining Features: 17 | AUC=0.9980 | ACC=0.9859
Remaining Features: 16 | AUC=0.9981 | ACC=0.9841
Remaining Features: 15 | AUC=0.9984 | ACC=0.9859
Remaining Features: 14 | AUC=0.9983 | ACC=0.9865
Remaining Features: 13 | AUC=0.9981 | ACC=0.9841
Remaining Features: 12 | AUC=0.9980 | ACC=0.9853
Remaining Features: 11 | AUC=0.9977 | ACC=0.9835
Remaining Features: 10 | AUC=0.9971 | ACC=0.9841
Remaining Features: 9 | AUC=0.9961 | ACC=0.9756
Remaining Features: 8 | AUC=0.9915 | ACC=0.9523
Remaining Features: 7 | AUC=0.9909 | ACC=0.9566
Remaining Features: 6 | AUC=0.9898 | ACC=0.9536
Remaining Features: 5 | AUC=0.9868 | ACC=0.9480
Remaining Features: 4 | AUC=0.9844 | ACC=0.9511
Remaining Features: 3 | AUC=0.9528 | ACC=0.9028
Remaining Features: 2 | AUC=0.8215 | ACC=0.7897

XGB Best Subset

Features : 15
Best AUC : 0.9983
Best ACC : 0.9865

Evaluating learner: RF

Remaining Features: 53 | AUC=0.9965 | ACC=0.9762
Remaining Features: 52 | AUC=0.9966 | ACC=0.9774
Remaining Features: 51 | AUC=0.9972 | ACC=0.9786
Remaining Features: 50 | AUC=0.9970 | ACC=0.9762
Remaining Features: 49 | AUC=0.9966 | ACC=0.9756
Remaining Features: 48 | AUC=0.9965 | ACC=0.9774

Remaining Features: 47 | AUC=0.9968 | ACC=0.9774
Remaining Features: 46 | AUC=0.9964 | ACC=0.9762
Remaining Features: 45 | AUC=0.9967 | ACC=0.9762
Remaining Features: 44 | AUC=0.9968 | ACC=0.9762
Remaining Features: 43 | AUC=0.9969 | ACC=0.9756
Remaining Features: 42 | AUC=0.9964 | ACC=0.9737
Remaining Features: 41 | AUC=0.9969 | ACC=0.9780
Remaining Features: 40 | AUC=0.9965 | ACC=0.9774
Remaining Features: 39 | AUC=0.9966 | ACC=0.9762
Remaining Features: 38 | AUC=0.9967 | ACC=0.9780
Remaining Features: 37 | AUC=0.9970 | ACC=0.9780
Remaining Features: 36 | AUC=0.9966 | ACC=0.9762
Remaining Features: 35 | AUC=0.9967 | ACC=0.9774
Remaining Features: 34 | AUC=0.9970 | ACC=0.9755
Remaining Features: 33 | AUC=0.9966 | ACC=0.9756
Remaining Features: 32 | AUC=0.9967 | ACC=0.9762
Remaining Features: 31 | AUC=0.9971 | ACC=0.9780
Remaining Features: 30 | AUC=0.9970 | ACC=0.9780
Remaining Features: 29 | AUC=0.9968 | ACC=0.9774
Remaining Features: 28 | AUC=0.9970 | ACC=0.9792
Remaining Features: 27 | AUC=0.9967 | ACC=0.9756
Remaining Features: 26 | AUC=0.9963 | ACC=0.9749
Remaining Features: 25 | AUC=0.9963 | ACC=0.9749
Remaining Features: 24 | AUC=0.9960 | ACC=0.9743
Remaining Features: 23 | AUC=0.9967 | ACC=0.9774
Remaining Features: 22 | AUC=0.9967 | ACC=0.9725
Remaining Features: 21 | AUC=0.9964 | ACC=0.9756
Remaining Features: 20 | AUC=0.9959 | ACC=0.9713
Remaining Features: 19 | AUC=0.9959 | ACC=0.9749
Remaining Features: 18 | AUC=0.9960 | ACC=0.9743
Remaining Features: 17 | AUC=0.9962 | ACC=0.9762
Remaining Features: 16 | AUC=0.9961 | ACC=0.9762
Remaining Features: 15 | AUC=0.9960 | ACC=0.9749
Remaining Features: 14 | AUC=0.9961 | ACC=0.9737
Remaining Features: 13 | AUC=0.9956 | ACC=0.9743
Remaining Features: 12 | AUC=0.9947 | ACC=0.9694
Remaining Features: 11 | AUC=0.9941 | ACC=0.9688
Remaining Features: 10 | AUC=0.9946 | ACC=0.9658
Remaining Features: 9 | AUC=0.9931 | ACC=0.9597
Remaining Features: 8 | AUC=0.9911 | ACC=0.9591
Remaining Features: 7 | AUC=0.9920 | ACC=0.9603
Remaining Features: 6 | AUC=0.9878 | ACC=0.9444
Remaining Features: 5 | AUC=0.9870 | ACC=0.9425
Remaining Features: 4 | AUC=0.9861 | ACC=0.9383
Remaining Features: 3 | AUC=0.9829 | ACC=0.9395
Remaining Features: 2 | AUC=0.9695 | ACC=0.9156

RF Best Subset

Features : 29
Best AUC : 0.997
Best ACC : 0.9792

=====
bRFE Completed

=====
Selected Learner : XGB
Selected Features: 15
Best AUC : 0.9983
Best Accuracy : 0.9865
=====

Running ESOM...

Efficient SMOTE-based Oversampling Method (ESOM)

Original Class Distribution

Class 0: 905

Class 1: 731

Majority Class : 0

Minority Class : 1

Class Imbalance: 19.23%

Applying Adaptive SMOTE

k_neighbors = 5

Balanced Class Distribution

Class 0: 905

Class 1: 905

Generated Samples : 174

Running EOLF...

Evaluating : RF

Best AUC : 0.9971

Best ACC : 0.9762

Evaluating : XGB

Best AUC : 0.9986

Best ACC : 0.9862

Evaluating : LightGBM

Best AUC : 0.9988

Best ACC : 0.9884

EOLF Completed

	Model	AUC	Accuracy \
0	RF	0.997112	0.976243
1	XGB	0.998559	0.986188
2	LightGBM	0.998834	0.988398

BestParameters

0 {'max_depth': None, 'min_samples_leaf': 1, 'mi...

1 {'learning_rate': 0.1, 'max_depth': 5, 'n_esti...

2 {'learning_rate': 0.1, 'n_estimators': 200, 'n...

Selected Model : LightGBM

Best AUC : 0.9988

Best ACC : 0.9884

Training Final Model...

Fold Results

Model : LightGBM

Features : 15

Accuracy : 0.9725

AUC : 0.9912

Precision : 0.9524

Recall : 0.9877

Specificity : 0.9604
F1-score : 0.9697
PR-AUC : 0.9891

=====

Outer Fold 3

=====

Running bRFE...

=====

Best Recursive Feature Elimination (bRFE)

=====

Evaluating learner: XGB

Remaining Features: 53	AUC=0.9976	ACC=0.9829
Remaining Features: 52	AUC=0.9973	ACC=0.9835
Remaining Features: 51	AUC=0.9979	ACC=0.9841
Remaining Features: 50	AUC=0.9971	ACC=0.9835
Remaining Features: 49	AUC=0.9977	ACC=0.9841
Remaining Features: 48	AUC=0.9974	ACC=0.9835
Remaining Features: 47	AUC=0.9972	ACC=0.9823
Remaining Features: 46	AUC=0.9977	ACC=0.9841
Remaining Features: 45	AUC=0.9974	ACC=0.9829
Remaining Features: 44	AUC=0.9972	ACC=0.9859
Remaining Features: 43	AUC=0.9977	ACC=0.9835
Remaining Features: 42	AUC=0.9975	ACC=0.9841
Remaining Features: 41	AUC=0.9972	ACC=0.9853
Remaining Features: 40	AUC=0.9977	ACC=0.9835
Remaining Features: 39	AUC=0.9974	ACC=0.9841
Remaining Features: 38	AUC=0.9976	ACC=0.9811
Remaining Features: 37	AUC=0.9973	ACC=0.9835
Remaining Features: 36	AUC=0.9977	ACC=0.9847
Remaining Features: 35	AUC=0.9974	ACC=0.9835
Remaining Features: 34	AUC=0.9977	ACC=0.9823
Remaining Features: 33	AUC=0.9976	ACC=0.9823
Remaining Features: 32	AUC=0.9976	ACC=0.9841
Remaining Features: 31	AUC=0.9976	ACC=0.9835
Remaining Features: 30	AUC=0.9976	ACC=0.9847
Remaining Features: 29	AUC=0.9977	ACC=0.9811
Remaining Features: 28	AUC=0.9978	ACC=0.9847
Remaining Features: 27	AUC=0.9976	ACC=0.9835
Remaining Features: 26	AUC=0.9980	ACC=0.9835
Remaining Features: 25	AUC=0.9977	ACC=0.9853
Remaining Features: 24	AUC=0.9975	ACC=0.9835
Remaining Features: 23	AUC=0.9977	ACC=0.9823
Remaining Features: 22	AUC=0.9975	ACC=0.9847
Remaining Features: 21	AUC=0.9978	ACC=0.9853
Remaining Features: 20	AUC=0.9979	ACC=0.9865
Remaining Features: 19	AUC=0.9977	ACC=0.9859
Remaining Features: 18	AUC=0.9978	ACC=0.9866
Remaining Features: 17	AUC=0.9976	ACC=0.9853
Remaining Features: 16	AUC=0.9977	ACC=0.9841
Remaining Features: 15	AUC=0.9976	ACC=0.9847
Remaining Features: 14	AUC=0.9968	ACC=0.9804
Remaining Features: 13	AUC=0.9970	ACC=0.9823
Remaining Features: 12	AUC=0.9972	ACC=0.9835
Remaining Features: 11	AUC=0.9967	ACC=0.9798
Remaining Features: 10	AUC=0.9968	ACC=0.9768
Remaining Features: 9	AUC=0.9969	ACC=0.9811
Remaining Features: 8	AUC=0.9958	ACC=0.9737
Remaining Features: 7	AUC=0.9913	ACC=0.9597
Remaining Features: 6	AUC=0.9883	ACC=0.9535
Remaining Features: 5	AUC=0.9876	ACC=0.9578
Remaining Features: 4	AUC=0.9812	ACC=0.9554
Remaining Features: 3	AUC=0.9536	ACC=0.9071

Remaining Features: 2 | AUC=0.8210 | ACC=0.7897

XGB Best Subset

Features : 19

Best AUC : 0.9978

Best ACC : 0.9866

Evaluating learner: RF

Remaining Features: 53 | AUC=0.9963 | ACC=0.9755
Remaining Features: 52 | AUC=0.9965 | ACC=0.9749
Remaining Features: 51 | AUC=0.9971 | ACC=0.9792
Remaining Features: 50 | AUC=0.9968 | ACC=0.9768
Remaining Features: 49 | AUC=0.9965 | ACC=0.9749
Remaining Features: 48 | AUC=0.9965 | ACC=0.9737
Remaining Features: 47 | AUC=0.9968 | ACC=0.9755
Remaining Features: 46 | AUC=0.9965 | ACC=0.9768
Remaining Features: 45 | AUC=0.9965 | ACC=0.9713
Remaining Features: 44 | AUC=0.9965 | ACC=0.9768
Remaining Features: 43 | AUC=0.9963 | ACC=0.9731
Remaining Features: 42 | AUC=0.9966 | ACC=0.9762
Remaining Features: 41 | AUC=0.9969 | ACC=0.9768
Remaining Features: 40 | AUC=0.9967 | ACC=0.9731
Remaining Features: 39 | AUC=0.9966 | ACC=0.9743
Remaining Features: 38 | AUC=0.9967 | ACC=0.9755
Remaining Features: 37 | AUC=0.9968 | ACC=0.9755
Remaining Features: 36 | AUC=0.9969 | ACC=0.9762
Remaining Features: 35 | AUC=0.9967 | ACC=0.9762
Remaining Features: 34 | AUC=0.9971 | ACC=0.9774
Remaining Features: 33 | AUC=0.9970 | ACC=0.9725
Remaining Features: 32 | AUC=0.9965 | ACC=0.9737
Remaining Features: 31 | AUC=0.9967 | ACC=0.9762
Remaining Features: 30 | AUC=0.9973 | ACC=0.9774
Remaining Features: 29 | AUC=0.9965 | ACC=0.9749
Remaining Features: 28 | AUC=0.9968 | ACC=0.9768
Remaining Features: 27 | AUC=0.9968 | ACC=0.9792
Remaining Features: 26 | AUC=0.9965 | ACC=0.9743
Remaining Features: 25 | AUC=0.9964 | ACC=0.9743
Remaining Features: 24 | AUC=0.9961 | ACC=0.9731
Remaining Features: 23 | AUC=0.9960 | ACC=0.9743
Remaining Features: 22 | AUC=0.9966 | ACC=0.9737
Remaining Features: 21 | AUC=0.9963 | ACC=0.9762
Remaining Features: 20 | AUC=0.9960 | ACC=0.9737
Remaining Features: 19 | AUC=0.9960 | ACC=0.9762
Remaining Features: 18 | AUC=0.9956 | ACC=0.9743
Remaining Features: 17 | AUC=0.9955 | ACC=0.9731
Remaining Features: 16 | AUC=0.9952 | ACC=0.9737
Remaining Features: 15 | AUC=0.9957 | ACC=0.9737
Remaining Features: 14 | AUC=0.9960 | ACC=0.9737
Remaining Features: 13 | AUC=0.9957 | ACC=0.9719
Remaining Features: 12 | AUC=0.9947 | ACC=0.9713
Remaining Features: 11 | AUC=0.9941 | ACC=0.9670
Remaining Features: 10 | AUC=0.9945 | ACC=0.9688
Remaining Features: 9 | AUC=0.9935 | ACC=0.9633
Remaining Features: 8 | AUC=0.9892 | ACC=0.9499
Remaining Features: 7 | AUC=0.9868 | ACC=0.9444
Remaining Features: 6 | AUC=0.9857 | ACC=0.9425
Remaining Features: 5 | AUC=0.9849 | ACC=0.9419
Remaining Features: 4 | AUC=0.9861 | ACC=0.9474
Remaining Features: 3 | AUC=0.9806 | ACC=0.9395
Remaining Features: 2 | AUC=0.9680 | ACC=0.9120

RF Best Subset

Features : 28
Best AUC : 0.9968
Best ACC : 0.9792

=====

bRFE Completed

=====

Selected Learner : XGB
Selected Features: 19
Best AUC : 0.9978
Best Accuracy : 0.9866

=====

Running ESOM...

=====

Efficient SMOTE-based Oversampling Method (ESOM)

=====

Original Class Distribution
Class 0: 905
Class 1: 731

Majority Class : 0
Minority Class : 1
Class Imbalance: 19.23%

Applying Adaptive SMOTE
k_neighbors = 5

Balanced Class Distribution
Class 0: 905
Class 1: 905

Generated Samples : 174

=====

Running EOLF...

=====

Evaluating : RF
Best AUC : 0.9977
Best ACC : 0.9807

=====

Evaluating : XGB
Best AUC : 0.9984
Best ACC : 0.984

=====

Evaluating : LightGBM
Best AUC : 0.9991
Best ACC : 0.9878

=====

EOLF Completed

=====

	Model	AUC	Accuracy \
0	RF	0.997668	0.980663
1	XGB	0.998425	0.983978
2	LightGBM	0.999121	0.987845

BestParameters
0 {'max_depth': None, 'min_samples_leaf': 1, 'mi...

- 1 {'learning_rate': 0.1, 'max_depth': 3, 'n_esti...
- 2 {'learning_rate': 0.1, 'n_estimators': 200, 'n...

Selected Model : LightGBM

Best AUC : 0.9991

Best ACC : 0.9878

Training Final Model...

Fold Results

Model : LightGBM

Features : 19

Accuracy : 0.9890

AUC : 0.9998

Precision : 1.0000

Recall : 0.9753

Specificity : 1.0000

F1-score : 0.9875

PR-AUC : 0.9997

=====
Outer Fold 4
=====

Running bRFE...

=====
Best Recursive Feature Elimination (bRFE)
=====

Evaluating learner: XGB

Remaining Features: 53	AUC=0.9985	ACC=0.9859
Remaining Features: 52	AUC=0.9983	ACC=0.9823
Remaining Features: 51	AUC=0.9986	ACC=0.9835
Remaining Features: 50	AUC=0.9984	ACC=0.9859
Remaining Features: 49	AUC=0.9984	ACC=0.9866
Remaining Features: 48	AUC=0.9984	ACC=0.9859
Remaining Features: 47	AUC=0.9983	ACC=0.9823
Remaining Features: 46	AUC=0.9985	ACC=0.9859
Remaining Features: 45	AUC=0.9983	ACC=0.9853
Remaining Features: 44	AUC=0.9983	ACC=0.9841
Remaining Features: 43	AUC=0.9985	ACC=0.9829
Remaining Features: 42	AUC=0.9983	ACC=0.9841
Remaining Features: 41	AUC=0.9983	ACC=0.9866
Remaining Features: 40	AUC=0.9985	ACC=0.9872
Remaining Features: 39	AUC=0.9984	ACC=0.9853
Remaining Features: 38	AUC=0.9984	ACC=0.9835
Remaining Features: 37	AUC=0.9985	ACC=0.9829
Remaining Features: 36	AUC=0.9983	ACC=0.9859
Remaining Features: 35	AUC=0.9984	ACC=0.9866
Remaining Features: 34	AUC=0.9984	ACC=0.9847
Remaining Features: 33	AUC=0.9986	ACC=0.9872
Remaining Features: 32	AUC=0.9985	ACC=0.9847
Remaining Features: 31	AUC=0.9984	ACC=0.9841
Remaining Features: 30	AUC=0.9984	ACC=0.9835
Remaining Features: 29	AUC=0.9984	ACC=0.9859
Remaining Features: 28	AUC=0.9984	ACC=0.9841
Remaining Features: 27	AUC=0.9984	ACC=0.9829
Remaining Features: 26	AUC=0.9986	ACC=0.9853
Remaining Features: 25	AUC=0.9985	ACC=0.9853
Remaining Features: 24	AUC=0.9985	ACC=0.9859
Remaining Features: 23	AUC=0.9984	ACC=0.9853
Remaining Features: 22	AUC=0.9985	ACC=0.9847
Remaining Features: 21	AUC=0.9985	ACC=0.9853
Remaining Features: 20	AUC=0.9982	ACC=0.9866
Remaining Features: 19	AUC=0.9983	ACC=0.9866

Remaining Features: 18	AUC=0.9986	ACC=0.9841
Remaining Features: 17	AUC=0.9987	ACC=0.9847
Remaining Features: 16	AUC=0.9986	ACC=0.9847
Remaining Features: 15	AUC=0.9988	ACC=0.9847
Remaining Features: 14	AUC=0.9983	ACC=0.9841
Remaining Features: 13	AUC=0.9984	ACC=0.9817
Remaining Features: 12	AUC=0.9985	ACC=0.9829
Remaining Features: 11	AUC=0.9981	ACC=0.9792
Remaining Features: 10	AUC=0.9927	ACC=0.9523
Remaining Features: 9	AUC=0.9914	ACC=0.9590
Remaining Features: 8	AUC=0.9905	ACC=0.9548
Remaining Features: 7	AUC=0.9896	ACC=0.9529
Remaining Features: 6	AUC=0.9885	ACC=0.9450
Remaining Features: 5	AUC=0.9843	ACC=0.9523
Remaining Features: 4	AUC=0.9592	ACC=0.9101
Remaining Features: 3	AUC=0.8383	ACC=0.7910
Remaining Features: 2	AUC=0.8156	ACC=0.7855

XGB Best Subset

Features : 20
Best AUC : 0.9983
Best ACC : 0.9866

Evaluating learner: RF

Remaining Features: 53	AUC=0.9970	ACC=0.9749
Remaining Features: 52	AUC=0.9970	ACC=0.9731
Remaining Features: 51	AUC=0.9975	ACC=0.9743
Remaining Features: 50	AUC=0.9970	ACC=0.9700
Remaining Features: 49	AUC=0.9968	ACC=0.9725
Remaining Features: 48	AUC=0.9971	ACC=0.9731
Remaining Features: 47	AUC=0.9972	ACC=0.9749
Remaining Features: 46	AUC=0.9974	ACC=0.9737
Remaining Features: 45	AUC=0.9967	ACC=0.9737
Remaining Features: 44	AUC=0.9970	ACC=0.9731
Remaining Features: 43	AUC=0.9971	ACC=0.9743
Remaining Features: 42	AUC=0.9971	ACC=0.9749
Remaining Features: 41	AUC=0.9970	ACC=0.9774
Remaining Features: 40	AUC=0.9967	ACC=0.9780
Remaining Features: 39	AUC=0.9974	ACC=0.9786
Remaining Features: 38	AUC=0.9975	ACC=0.9792
Remaining Features: 37	AUC=0.9974	ACC=0.9798
Remaining Features: 36	AUC=0.9973	ACC=0.9755
Remaining Features: 35	AUC=0.9974	ACC=0.9762
Remaining Features: 34	AUC=0.9978	ACC=0.9792
Remaining Features: 33	AUC=0.9972	ACC=0.9762
Remaining Features: 32	AUC=0.9973	ACC=0.9762
Remaining Features: 31	AUC=0.9974	ACC=0.9786
Remaining Features: 30	AUC=0.9973	ACC=0.9768
Remaining Features: 29	AUC=0.9967	ACC=0.9731
Remaining Features: 28	AUC=0.9964	ACC=0.9755
Remaining Features: 27	AUC=0.9965	ACC=0.9743
Remaining Features: 26	AUC=0.9966	ACC=0.9737
Remaining Features: 25	AUC=0.9966	ACC=0.9737
Remaining Features: 24	AUC=0.9965	ACC=0.9737
Remaining Features: 23	AUC=0.9967	ACC=0.9743
Remaining Features: 22	AUC=0.9969	ACC=0.9749
Remaining Features: 21	AUC=0.9970	ACC=0.9743
Remaining Features: 20	AUC=0.9964	ACC=0.9737
Remaining Features: 19	AUC=0.9968	ACC=0.9755
Remaining Features: 18	AUC=0.9967	ACC=0.9737
Remaining Features: 17	AUC=0.9964	ACC=0.9713
Remaining Features: 16	AUC=0.9961	ACC=0.9725
Remaining Features: 15	AUC=0.9962	ACC=0.9762
Remaining Features: 14	AUC=0.9968	ACC=0.9743

Remaining Features: 13 | AUC=0.9960 | ACC=0.9725
Remaining Features: 12 | AUC=0.9960 | ACC=0.9731
Remaining Features: 11 | AUC=0.9945 | ACC=0.9664
Remaining Features: 10 | AUC=0.9946 | ACC=0.9639
Remaining Features: 9 | AUC=0.9933 | ACC=0.9578
Remaining Features: 8 | AUC=0.9942 | ACC=0.9645
Remaining Features: 7 | AUC=0.9896 | ACC=0.9523
Remaining Features: 6 | AUC=0.9886 | ACC=0.9511
Remaining Features: 5 | AUC=0.9887 | ACC=0.9505
Remaining Features: 4 | AUC=0.9851 | ACC=0.9419
Remaining Features: 3 | AUC=0.9817 | ACC=0.9389
Remaining Features: 2 | AUC=0.9642 | ACC=0.9089

RF Best Subset

Features : 32
Best AUC : 0.9974
Best ACC : 0.9786

=====

bRFE Completed

=====

Selected Learner : XGB
Selected Features: 20
Best AUC : 0.9983
Best Accuracy : 0.9866
=====

Running ESOM...

=====

Efficient SMOTE-based Oversampling Method (ESOM)

=====

Original Class Distribution
Class 0: 905
Class 1: 731

Majority Class : 0
Minority Class : 1
Class Imbalance: 19.23%

Applying Adaptive SMOTE
k_neighbors = 5

Balanced Class Distribution
Class 0: 905
Class 1: 905

Generated Samples : 174
=====

Running EOLF...

=====

Evaluating : RF
Best AUC : 0.9977
Best ACC : 0.9834

=====

Evaluating : XGB
Best AUC : 0.9988
Best ACC : 0.9873

```
=====
Evaluating : LightGBM
Best AUC : 0.9987
Best ACC : 0.9867
```

```
=====
EOLF Completed
=====
```

```
Model    AUC  Accuracy \
0    RF  0.997680  0.983425
1    XGB  0.998822  0.987293
2  LightGBM  0.998700  0.986740
```

```
BestParameters
0 {'max_depth': None, 'min_samples_leaf': 1, 'mi...
1 {'learning_rate': 0.1, 'max_depth': 3, 'n_esti...
2 {'learning_rate': 0.1, 'n_estimators': 200, 'n...
```

```
Selected Model : XGB
Best AUC : 0.9988
Best ACC : 0.9873
```

```
Training Final Model...
```

```
Fold Results
Model      : XGB
Features   : 20
Accuracy   : 0.9890
AUC        : 0.9989
Precision  : 0.9877
Recall     : 0.9877
Specificity : 0.9901
F1-score   : 0.9877
PR-AUC     : 0.9987
```

```
=====
Outer Fold 5
=====
```

```
Running bRFE...
```

```
=====
Best Recursive Feature Elimination (bRFE)
=====
```

```
Evaluating learner: XGB
Remaining Features: 53 | AUC=0.9977 | ACC=0.9853
Remaining Features: 52 | AUC=0.9974 | ACC=0.9859
Remaining Features: 51 | AUC=0.9979 | ACC=0.9829
Remaining Features: 50 | AUC=0.9978 | ACC=0.9859
Remaining Features: 49 | AUC=0.9979 | ACC=0.9847
Remaining Features: 48 | AUC=0.9976 | ACC=0.9847
Remaining Features: 47 | AUC=0.9974 | ACC=0.9853
Remaining Features: 46 | AUC=0.9978 | ACC=0.9847
Remaining Features: 45 | AUC=0.9975 | ACC=0.9853
Remaining Features: 44 | AUC=0.9975 | ACC=0.9853
Remaining Features: 43 | AUC=0.9975 | ACC=0.9835
Remaining Features: 42 | AUC=0.9978 | ACC=0.9835
Remaining Features: 41 | AUC=0.9978 | ACC=0.9853
Remaining Features: 40 | AUC=0.9976 | ACC=0.9847
Remaining Features: 39 | AUC=0.9977 | ACC=0.9829
Remaining Features: 38 | AUC=0.9980 | ACC=0.9865
Remaining Features: 37 | AUC=0.9979 | ACC=0.9841
Remaining Features: 36 | AUC=0.9975 | ACC=0.9841
Remaining Features: 35 | AUC=0.9981 | ACC=0.9841
```

Remaining Features: 34	AUC=0.9978	ACC=0.9847
Remaining Features: 33	AUC=0.9980	ACC=0.9853
Remaining Features: 32	AUC=0.9981	ACC=0.9853
Remaining Features: 31	AUC=0.9980	ACC=0.9841
Remaining Features: 30	AUC=0.9983	ACC=0.9865
Remaining Features: 29	AUC=0.9981	ACC=0.9859
Remaining Features: 28	AUC=0.9983	ACC=0.9853
Remaining Features: 27	AUC=0.9983	ACC=0.9853
Remaining Features: 26	AUC=0.9984	ACC=0.9853
Remaining Features: 25	AUC=0.9981	ACC=0.9835
Remaining Features: 24	AUC=0.9982	ACC=0.9853
Remaining Features: 23	AUC=0.9982	ACC=0.9859
Remaining Features: 22	AUC=0.9982	ACC=0.9859
Remaining Features: 21	AUC=0.9982	ACC=0.9859
Remaining Features: 20	AUC=0.9983	ACC=0.9853
Remaining Features: 19	AUC=0.9984	ACC=0.9872
Remaining Features: 18	AUC=0.9983	ACC=0.9865
Remaining Features: 17	AUC=0.9977	ACC=0.9859
Remaining Features: 16	AUC=0.9976	ACC=0.9859
Remaining Features: 15	AUC=0.9980	ACC=0.9865
Remaining Features: 14	AUC=0.9975	ACC=0.9810
Remaining Features: 13	AUC=0.9974	ACC=0.9823
Remaining Features: 12	AUC=0.9966	ACC=0.9817
Remaining Features: 11	AUC=0.9939	ACC=0.9682
Remaining Features: 10	AUC=0.9949	ACC=0.9768
Remaining Features: 9	AUC=0.9896	ACC=0.9609
Remaining Features: 8	AUC=0.9915	ACC=0.9621
Remaining Features: 7	AUC=0.9895	ACC=0.9523
Remaining Features: 6	AUC=0.9891	ACC=0.9529
Remaining Features: 5	AUC=0.9888	ACC=0.9548
Remaining Features: 4	AUC=0.9841	ACC=0.9535
Remaining Features: 3	AUC=0.9529	ACC=0.9028
Remaining Features: 2	AUC=0.8180	ACC=0.7897

XGB Best Subset

Features : 16
Best AUC : 0.998
Best ACC : 0.9865

Evaluating learner: RF

Remaining Features: 53	AUC=0.9963	ACC=0.9749
Remaining Features: 52	AUC=0.9962	ACC=0.9737
Remaining Features: 51	AUC=0.9965	ACC=0.9762
Remaining Features: 50	AUC=0.9965	ACC=0.9762
Remaining Features: 49	AUC=0.9964	ACC=0.9731
Remaining Features: 48	AUC=0.9968	ACC=0.9755
Remaining Features: 47	AUC=0.9963	ACC=0.9749
Remaining Features: 46	AUC=0.9966	ACC=0.9762
Remaining Features: 45	AUC=0.9962	ACC=0.9743
Remaining Features: 44	AUC=0.9967	ACC=0.9725
Remaining Features: 43	AUC=0.9964	ACC=0.9768
Remaining Features: 42	AUC=0.9965	ACC=0.9719
Remaining Features: 41	AUC=0.9963	ACC=0.9731
Remaining Features: 40	AUC=0.9961	ACC=0.9780
Remaining Features: 39	AUC=0.9967	ACC=0.9780
Remaining Features: 38	AUC=0.9967	ACC=0.9774
Remaining Features: 37	AUC=0.9967	ACC=0.9780
Remaining Features: 36	AUC=0.9966	ACC=0.9737
Remaining Features: 35	AUC=0.9967	ACC=0.9780
Remaining Features: 34	AUC=0.9970	ACC=0.9774
Remaining Features: 33	AUC=0.9970	ACC=0.9780
Remaining Features: 32	AUC=0.9967	ACC=0.9749
Remaining Features: 31	AUC=0.9966	ACC=0.9737
Remaining Features: 30	AUC=0.9964	ACC=0.9768

Remaining Features: 29 | AUC=0.9964 | ACC=0.9755
Remaining Features: 28 | AUC=0.9962 | ACC=0.9768
Remaining Features: 27 | AUC=0.9963 | ACC=0.9768
Remaining Features: 26 | AUC=0.9960 | ACC=0.9737
Remaining Features: 25 | AUC=0.9960 | ACC=0.9749
Remaining Features: 24 | AUC=0.9962 | ACC=0.9780
Remaining Features: 23 | AUC=0.9961 | ACC=0.9743
Remaining Features: 22 | AUC=0.9962 | ACC=0.9737
Remaining Features: 21 | AUC=0.9958 | ACC=0.9719
Remaining Features: 20 | AUC=0.9956 | ACC=0.9743
Remaining Features: 19 | AUC=0.9960 | ACC=0.9743
Remaining Features: 18 | AUC=0.9960 | ACC=0.9749
Remaining Features: 17 | AUC=0.9964 | ACC=0.9755
Remaining Features: 16 | AUC=0.9957 | ACC=0.9743
Remaining Features: 15 | AUC=0.9957 | ACC=0.9743
Remaining Features: 14 | AUC=0.9960 | ACC=0.9749
Remaining Features: 13 | AUC=0.9960 | ACC=0.9755
Remaining Features: 12 | AUC=0.9949 | ACC=0.9719
Remaining Features: 11 | AUC=0.9939 | ACC=0.9664
Remaining Features: 10 | AUC=0.9943 | ACC=0.9700
Remaining Features: 9 | AUC=0.9929 | ACC=0.9670
Remaining Features: 8 | AUC=0.9897 | ACC=0.9542
Remaining Features: 7 | AUC=0.9869 | ACC=0.9511
Remaining Features: 6 | AUC=0.9866 | ACC=0.9462
Remaining Features: 5 | AUC=0.9870 | ACC=0.9474
Remaining Features: 4 | AUC=0.9874 | ACC=0.9450
Remaining Features: 3 | AUC=0.9805 | ACC=0.9395
Remaining Features: 2 | AUC=0.9716 | ACC=0.9169

RF Best Subset

Features : 25
Best AUC : 0.9962
Best ACC : 0.978

=====

bRFE Completed

=====

Selected Learner : XGB
Selected Features: 16
Best AUC : 0.9980
Best Accuracy : 0.9865

=====

Running ESOM...

=====

Efficient SMOTE-based Oversampling Method (ESOM)

=====

Original Class Distribution

Class 0: 905

Class 1: 731

Majority Class : 0

Minority Class : 1

Class Imbalance: 19.23%

Applying Adaptive SMOTE

k_neighbors = 5

Balanced Class Distribution

Class 0: 905

Class 1: 905

Generated Samples : 174

Running EOLF...

Evaluating : RF
Best AUC : 0.9967
Best ACC : 0.9762

Evaluating : XGB
Best AUC : 0.9985
Best ACC : 0.9856

Evaluating : LightGBM
Best AUC : 0.9987
Best ACC : 0.9867

EOLF Completed

	Model	AUC	Accuracy \
0	RF	0.996676	0.976243
1	XGB	0.998498	0.985635
2	LightGBM	0.998700	0.986740

BestParameters
0 {'max_depth': None, 'min_samples_leaf': 1, 'mi...
1 {'learning_rate': 0.1, 'max_depth': 3, 'n_esti...
2 {'learning_rate': 0.1, 'n_estimators': 200, 'n...

Selected Model : LightGBM
Best AUC : 0.9987
Best ACC : 0.9867

Training Final Model...

Fold Results
Model : LightGBM
Features : 16
Accuracy : 0.9780
AUC : 0.9966
Precision : 0.9753
Recall : 0.9753
Specificity : 0.9802
F1-score : 0.9753
PR-AUC : 0.9951

Outer Fold 6

Running bRFE...

Best Recursive Feature Elimination (bRFE)

Evaluating learner: XGB
Remaining Features: 53 | AUC=0.9978 | ACC=0.9847
Remaining Features: 52 | AUC=0.9979 | ACC=0.9853
Remaining Features: 51 | AUC=0.9978 | ACC=0.9829

Remaining Features: 50	AUC=0.9980	ACC=0.9829
Remaining Features: 49	AUC=0.9980	ACC=0.9841
Remaining Features: 48	AUC=0.9981	ACC=0.9859
Remaining Features: 47	AUC=0.9979	ACC=0.9841
Remaining Features: 46	AUC=0.9978	ACC=0.9823
Remaining Features: 45	AUC=0.9979	ACC=0.9829
Remaining Features: 44	AUC=0.9980	ACC=0.9835
Remaining Features: 43	AUC=0.9977	ACC=0.9841
Remaining Features: 42	AUC=0.9973	ACC=0.9835
Remaining Features: 41	AUC=0.9981	ACC=0.9847
Remaining Features: 40	AUC=0.9977	ACC=0.9811
Remaining Features: 39	AUC=0.9979	ACC=0.9829
Remaining Features: 38	AUC=0.9979	ACC=0.9786
Remaining Features: 37	AUC=0.9979	ACC=0.9804
Remaining Features: 36	AUC=0.9977	ACC=0.9817
Remaining Features: 35	AUC=0.9979	ACC=0.9817
Remaining Features: 34	AUC=0.9978	ACC=0.9859
Remaining Features: 33	AUC=0.9977	ACC=0.9847
Remaining Features: 32	AUC=0.9978	ACC=0.9841
Remaining Features: 31	AUC=0.9981	ACC=0.9829
Remaining Features: 30	AUC=0.9977	ACC=0.9829
Remaining Features: 29	AUC=0.9981	ACC=0.9853
Remaining Features: 28	AUC=0.9980	ACC=0.9847
Remaining Features: 27	AUC=0.9981	ACC=0.9847
Remaining Features: 26	AUC=0.9983	ACC=0.9835
Remaining Features: 25	AUC=0.9981	ACC=0.9810
Remaining Features: 24	AUC=0.9982	ACC=0.9835
Remaining Features: 23	AUC=0.9981	ACC=0.9835
Remaining Features: 22	AUC=0.9983	ACC=0.9817
Remaining Features: 21	AUC=0.9981	ACC=0.9841
Remaining Features: 20	AUC=0.9980	ACC=0.9829
Remaining Features: 19	AUC=0.9975	ACC=0.9847
Remaining Features: 18	AUC=0.9982	ACC=0.9859
Remaining Features: 17	AUC=0.9976	ACC=0.9847
Remaining Features: 16	AUC=0.9979	ACC=0.9841
Remaining Features: 15	AUC=0.9974	ACC=0.9847
Remaining Features: 14	AUC=0.9969	ACC=0.9847
Remaining Features: 13	AUC=0.9975	ACC=0.9823
Remaining Features: 12	AUC=0.9976	ACC=0.9823
Remaining Features: 11	AUC=0.9971	ACC=0.9798
Remaining Features: 10	AUC=0.9976	ACC=0.9804
Remaining Features: 9	AUC=0.9973	ACC=0.9804
Remaining Features: 8	AUC=0.9971	ACC=0.9798
Remaining Features: 7	AUC=0.9957	ACC=0.9749
Remaining Features: 6	AUC=0.9950	ACC=0.9749
Remaining Features: 5	AUC=0.9954	ACC=0.9768
Remaining Features: 4	AUC=0.9849	ACC=0.9584
Remaining Features: 3	AUC=0.9561	ACC=0.8985
Remaining Features: 2	AUC=0.8206	ACC=0.7891

XGB Best Subset

Features : 19
Best AUC : 0.9982
Best ACC : 0.9859

Evaluating learner: RF

Remaining Features: 53	AUC=0.9966	ACC=0.9707
Remaining Features: 52	AUC=0.9969	ACC=0.9756
Remaining Features: 51	AUC=0.9972	ACC=0.9774
Remaining Features: 50	AUC=0.9968	ACC=0.9731
Remaining Features: 49	AUC=0.9968	ACC=0.9774
Remaining Features: 48	AUC=0.9970	ACC=0.9762
Remaining Features: 47	AUC=0.9969	ACC=0.9768
Remaining Features: 46	AUC=0.9966	ACC=0.9713

Remaining Features: 45 | AUC=0.9965 | ACC=0.9719
Remaining Features: 44 | AUC=0.9971 | ACC=0.9725
Remaining Features: 43 | AUC=0.9972 | ACC=0.9749
Remaining Features: 42 | AUC=0.9972 | ACC=0.9755
Remaining Features: 41 | AUC=0.9973 | ACC=0.9762
Remaining Features: 40 | AUC=0.9971 | ACC=0.9774
Remaining Features: 39 | AUC=0.9973 | ACC=0.9762
Remaining Features: 38 | AUC=0.9970 | ACC=0.9780
Remaining Features: 37 | AUC=0.9970 | ACC=0.9762
Remaining Features: 36 | AUC=0.9975 | ACC=0.9774
Remaining Features: 35 | AUC=0.9972 | ACC=0.9774
Remaining Features: 34 | AUC=0.9975 | ACC=0.9786
Remaining Features: 33 | AUC=0.9968 | ACC=0.9755
Remaining Features: 32 | AUC=0.9975 | ACC=0.9804
Remaining Features: 31 | AUC=0.9974 | ACC=0.9755
Remaining Features: 30 | AUC=0.9974 | ACC=0.9774
Remaining Features: 29 | AUC=0.9971 | ACC=0.9768
Remaining Features: 28 | AUC=0.9970 | ACC=0.9725
Remaining Features: 27 | AUC=0.9970 | ACC=0.9737
Remaining Features: 26 | AUC=0.9968 | ACC=0.9725
Remaining Features: 25 | AUC=0.9967 | ACC=0.9743
Remaining Features: 24 | AUC=0.9965 | ACC=0.9762
Remaining Features: 23 | AUC=0.9966 | ACC=0.9762
Remaining Features: 22 | AUC=0.9970 | ACC=0.9792
Remaining Features: 21 | AUC=0.9966 | ACC=0.9749
Remaining Features: 20 | AUC=0.9965 | ACC=0.9743
Remaining Features: 19 | AUC=0.9967 | ACC=0.9762
Remaining Features: 18 | AUC=0.9964 | ACC=0.9731
Remaining Features: 17 | AUC=0.9966 | ACC=0.9755
Remaining Features: 16 | AUC=0.9964 | ACC=0.9749
Remaining Features: 15 | AUC=0.9962 | ACC=0.9743
Remaining Features: 14 | AUC=0.9963 | ACC=0.9737
Remaining Features: 13 | AUC=0.9955 | ACC=0.9707
Remaining Features: 12 | AUC=0.9945 | ACC=0.9700
Remaining Features: 11 | AUC=0.9938 | ACC=0.9664
Remaining Features: 10 | AUC=0.9950 | ACC=0.9664
Remaining Features: 9 | AUC=0.9938 | ACC=0.9646
Remaining Features: 8 | AUC=0.9911 | ACC=0.9548
Remaining Features: 7 | AUC=0.9858 | ACC=0.9444
Remaining Features: 6 | AUC=0.9850 | ACC=0.9432
Remaining Features: 5 | AUC=0.9850 | ACC=0.9432
Remaining Features: 4 | AUC=0.9735 | ACC=0.9248
Remaining Features: 3 | AUC=0.9746 | ACC=0.9260
Remaining Features: 2 | AUC=0.9667 | ACC=0.9095

RF Best Subset

Features : 33
Best AUC : 0.9975
Best ACC : 0.9804

=====
bRFE Completed

=====
Selected Learner : XGB
Selected Features: 19
Best AUC : 0.9982
Best Accuracy : 0.9859
=====

Running ESOM...

=====
Efficient SMOTE-based Oversampling Method (ESOM)

=====

Original Class Distribution

Class 0: 905

Class 1: 731

Majority Class : 0

Minority Class : 1

Class Imbalance: 19.23%

Applying Adaptive SMOTE

k_neighbors = 5

Balanced Class Distribution

Class 0: 905

Class 1: 905

Generated Samples : 174

=====

Running EOLF...

=====

Evaluating : RF

Best AUC : 0.9972

Best ACC : 0.9779

=====

Evaluating : XGB

Best AUC : 0.998

Best ACC : 0.9867

=====

Evaluating : LightGBM

Best AUC : 0.9982

Best ACC : 0.9856

=====

EOLF Completed

=====

	Model	AUC	Accuracy \
0	RF	0.997161	0.977901
1	XGB	0.997967	0.986740
2	LightGBM	0.998217	0.985635

BestParameters

0 {'max_depth': None, 'min_samples_leaf': 1, 'mi...

1 {'learning_rate': 0.1, 'max_depth': 3, 'n_esti...

2 {'learning_rate': 0.1, 'n_estimators': 200, 'n...

Selected Model : LightGBM

Best AUC : 0.9982

Best ACC : 0.9856

Training Final Model...

Fold Results

Model : LightGBM

Features : 19

Accuracy : 0.9835

AUC : 0.9996

Precision : 0.9875

Recall : 0.9753

Specificity : 0.9901

F1-score : 0.9814

PR-AUC : 0.9995

Outer Fold 7

Running bRFE...

Best Recursive Feature Elimination (bRFE)

Evaluating learner: XGB

Remaining Features: 53	AUC=0.9972	ACC=0.9829
Remaining Features: 52	AUC=0.9973	ACC=0.9810
Remaining Features: 51	AUC=0.9974	ACC=0.9811
Remaining Features: 50	AUC=0.9978	ACC=0.9811
Remaining Features: 49	AUC=0.9977	ACC=0.9835
Remaining Features: 48	AUC=0.9973	ACC=0.9811
Remaining Features: 47	AUC=0.9970	ACC=0.9823
Remaining Features: 46	AUC=0.9974	ACC=0.9804
Remaining Features: 45	AUC=0.9977	ACC=0.9829
Remaining Features: 44	AUC=0.9973	ACC=0.9823
Remaining Features: 43	AUC=0.9978	ACC=0.9829
Remaining Features: 42	AUC=0.9975	ACC=0.9817
Remaining Features: 41	AUC=0.9976	ACC=0.9823
Remaining Features: 40	AUC=0.9976	ACC=0.9835
Remaining Features: 39	AUC=0.9975	ACC=0.9804
Remaining Features: 38	AUC=0.9975	ACC=0.9811
Remaining Features: 37	AUC=0.9974	ACC=0.9823
Remaining Features: 36	AUC=0.9976	ACC=0.9829
Remaining Features: 35	AUC=0.9976	ACC=0.9829
Remaining Features: 34	AUC=0.9977	ACC=0.9829
Remaining Features: 33	AUC=0.9977	ACC=0.9829
Remaining Features: 32	AUC=0.9977	ACC=0.9823
Remaining Features: 31	AUC=0.9974	ACC=0.9829
Remaining Features: 30	AUC=0.9978	ACC=0.9835
Remaining Features: 29	AUC=0.9977	ACC=0.9798
Remaining Features: 28	AUC=0.9976	ACC=0.9823
Remaining Features: 27	AUC=0.9974	ACC=0.9792
Remaining Features: 26	AUC=0.9976	ACC=0.9841
Remaining Features: 25	AUC=0.9975	ACC=0.9829
Remaining Features: 24	AUC=0.9976	ACC=0.9817
Remaining Features: 23	AUC=0.9976	ACC=0.9823
Remaining Features: 22	AUC=0.9977	ACC=0.9823
Remaining Features: 21	AUC=0.9977	ACC=0.9804
Remaining Features: 20	AUC=0.9977	ACC=0.9841
Remaining Features: 19	AUC=0.9975	ACC=0.9829
Remaining Features: 18	AUC=0.9975	ACC=0.9823
Remaining Features: 17	AUC=0.9973	ACC=0.9865
Remaining Features: 16	AUC=0.9976	ACC=0.9865
Remaining Features: 15	AUC=0.9982	ACC=0.9829
Remaining Features: 14	AUC=0.9977	ACC=0.9829
Remaining Features: 13	AUC=0.9976	ACC=0.9823
Remaining Features: 12	AUC=0.9971	ACC=0.9792
Remaining Features: 11	AUC=0.9970	ACC=0.9798
Remaining Features: 10	AUC=0.9971	ACC=0.9804
Remaining Features: 9	AUC=0.9972	ACC=0.9804
Remaining Features: 8	AUC=0.9959	ACC=0.9719
Remaining Features: 7	AUC=0.9958	ACC=0.9743
Remaining Features: 6	AUC=0.9953	ACC=0.9700
Remaining Features: 5	AUC=0.9954	ACC=0.9713
Remaining Features: 4	AUC=0.9798	ACC=0.9480
Remaining Features: 3	AUC=0.9530	ACC=0.9034
Remaining Features: 2	AUC=0.8220	ACC=0.7916

XGB Best Subset

Features : 17
Best AUC : 0.9976
Best ACC : 0.9865

Evaluating learner: RF

Remaining Features: 53	AUC=0.9966	ACC=0.9713
Remaining Features: 52	AUC=0.9964	ACC=0.9731
Remaining Features: 51	AUC=0.9967	ACC=0.9749
Remaining Features: 50	AUC=0.9967	ACC=0.9731
Remaining Features: 49	AUC=0.9968	ACC=0.9749
Remaining Features: 48	AUC=0.9968	ACC=0.9707
Remaining Features: 47	AUC=0.9967	ACC=0.9719
Remaining Features: 46	AUC=0.9967	ACC=0.9725
Remaining Features: 45	AUC=0.9968	ACC=0.9682
Remaining Features: 44	AUC=0.9967	ACC=0.9706
Remaining Features: 43	AUC=0.9969	ACC=0.9743
Remaining Features: 42	AUC=0.9964	ACC=0.9700
Remaining Features: 41	AUC=0.9972	ACC=0.9743
Remaining Features: 40	AUC=0.9966	ACC=0.9707
Remaining Features: 39	AUC=0.9968	ACC=0.9743
Remaining Features: 38	AUC=0.9971	ACC=0.9737
Remaining Features: 37	AUC=0.9971	ACC=0.9743
Remaining Features: 36	AUC=0.9969	ACC=0.9719
Remaining Features: 35	AUC=0.9969	ACC=0.9768
Remaining Features: 34	AUC=0.9970	ACC=0.9755
Remaining Features: 33	AUC=0.9972	ACC=0.9755
Remaining Features: 32	AUC=0.9965	ACC=0.9725
Remaining Features: 31	AUC=0.9968	ACC=0.9762
Remaining Features: 30	AUC=0.9968	ACC=0.9737
Remaining Features: 29	AUC=0.9965	ACC=0.9731
Remaining Features: 28	AUC=0.9962	ACC=0.9719
Remaining Features: 27	AUC=0.9961	ACC=0.9737
Remaining Features: 26	AUC=0.9961	ACC=0.9707
Remaining Features: 25	AUC=0.9961	ACC=0.9731
Remaining Features: 24	AUC=0.9961	ACC=0.9713
Remaining Features: 23	AUC=0.9961	ACC=0.9694
Remaining Features: 22	AUC=0.9961	ACC=0.9707
Remaining Features: 21	AUC=0.9962	ACC=0.9719
Remaining Features: 20	AUC=0.9956	ACC=0.9713
Remaining Features: 19	AUC=0.9960	ACC=0.9725
Remaining Features: 18	AUC=0.9957	ACC=0.9700
Remaining Features: 17	AUC=0.9956	ACC=0.9737
Remaining Features: 16	AUC=0.9960	ACC=0.9743
Remaining Features: 15	AUC=0.9960	ACC=0.9731
Remaining Features: 14	AUC=0.9960	ACC=0.9707
Remaining Features: 13	AUC=0.9960	ACC=0.9731
Remaining Features: 12	AUC=0.9955	ACC=0.9700
Remaining Features: 11	AUC=0.9943	ACC=0.9670
Remaining Features: 10	AUC=0.9947	ACC=0.9670
Remaining Features: 9	AUC=0.9926	ACC=0.9615
Remaining Features: 8	AUC=0.9914	ACC=0.9548
Remaining Features: 7	AUC=0.9860	ACC=0.9450
Remaining Features: 6	AUC=0.9865	ACC=0.9450
Remaining Features: 5	AUC=0.9758	ACC=0.9279
Remaining Features: 4	AUC=0.9750	ACC=0.9248
Remaining Features: 3	AUC=0.9746	ACC=0.9242
Remaining Features: 2	AUC=0.9674	ACC=0.9132

RF Best Subset

Features : 32
Best AUC : 0.9968

Best ACC : 0.9762

=====

bRFE Completed

=====

Selected Learner : XGB
Selected Features: 17
Best AUC : 0.9976
Best Accuracy : 0.9865

=====

Running ESOM...

=====

Efficient SMOTE-based Oversampling Method (ESOM)

=====

Original Class Distribution
Class 0: 906
Class 1: 730

Majority Class : 0
Minority Class : 1
Class Imbalance: 19.43%

Applying Adaptive SMOTE
k_neighbors = 5

Balanced Class Distribution
Class 0: 906
Class 1: 906

Generated Samples : 176

=====

Running EOLF...

=====

Evaluating : RF
Best AUC : 0.997
Best ACC : 0.9768

=====

Evaluating : XGB
Best AUC : 0.9984
Best ACC : 0.9873

=====

Evaluating : LightGBM
Best AUC : 0.9987
Best ACC : 0.9862

=====

=====

EOLF Completed

=====

	Model	AUC	Accuracy \
0	RF	0.997006	0.976817
1	XGB	0.998383	0.987300
2	LightGBM	0.998688	0.986198

BestParameters

0	{'max_depth': None, 'min_samples_leaf': 1, 'mi...
1	{'learning_rate': 0.1, 'max_depth': 3, 'n_esti...
2	{'learning_rate': 0.1, 'n_estimators': 200, 'n...

Selected Model : LightGBM
Best AUC : 0.9987
Best ACC : 0.9862

Training Final Model...

Fold Results

Model : LightGBM
Features : 17
Accuracy : 0.9835
AUC : 0.9995
Precision : 0.9877
Recall : 0.9756
Specificity : 0.9900
F1-score : 0.9816
PR-AUC : 0.9994

=====
Outer Fold 8
=====

Running bRFE...

=====
Best Recursive Feature Elimination (bRFE)
=====

Evaluating learner: XGB

Remaining Features: 53	AUC=0.9977	ACC=0.9841
Remaining Features: 52	AUC=0.9978	ACC=0.9847
Remaining Features: 51	AUC=0.9977	ACC=0.9835
Remaining Features: 50	AUC=0.9975	ACC=0.9847
Remaining Features: 49	AUC=0.9975	ACC=0.9847
Remaining Features: 48	AUC=0.9974	ACC=0.9829
Remaining Features: 47	AUC=0.9976	ACC=0.9835
Remaining Features: 46	AUC=0.9970	ACC=0.9823
Remaining Features: 45	AUC=0.9976	ACC=0.9847
Remaining Features: 44	AUC=0.9974	ACC=0.9853
Remaining Features: 43	AUC=0.9975	ACC=0.9829
Remaining Features: 42	AUC=0.9973	ACC=0.9835
Remaining Features: 41	AUC=0.9978	ACC=0.9841
Remaining Features: 40	AUC=0.9977	ACC=0.9859
Remaining Features: 39	AUC=0.9978	ACC=0.9872
Remaining Features: 38	AUC=0.9978	ACC=0.9847
Remaining Features: 37	AUC=0.9976	ACC=0.9847
Remaining Features: 36	AUC=0.9976	ACC=0.9866
Remaining Features: 35	AUC=0.9980	ACC=0.9847
Remaining Features: 34	AUC=0.9975	ACC=0.9847
Remaining Features: 33	AUC=0.9976	ACC=0.9847
Remaining Features: 32	AUC=0.9978	ACC=0.9835
Remaining Features: 31	AUC=0.9976	ACC=0.9853
Remaining Features: 30	AUC=0.9979	ACC=0.9835
Remaining Features: 29	AUC=0.9978	ACC=0.9859
Remaining Features: 28	AUC=0.9979	ACC=0.9835
Remaining Features: 27	AUC=0.9981	ACC=0.9847
Remaining Features: 26	AUC=0.9976	ACC=0.9859
Remaining Features: 25	AUC=0.9978	ACC=0.9865
Remaining Features: 24	AUC=0.9976	ACC=0.9853
Remaining Features: 23	AUC=0.9979	ACC=0.9835
Remaining Features: 22	AUC=0.9980	ACC=0.9841
Remaining Features: 21	AUC=0.9974	ACC=0.9841
Remaining Features: 20	AUC=0.9979	ACC=0.9847
Remaining Features: 19	AUC=0.9976	ACC=0.9859
Remaining Features: 18	AUC=0.9968	ACC=0.9847
Remaining Features: 17	AUC=0.9971	ACC=0.9835

Remaining Features: 16	AUC=0.9970	ACC=0.9835
Remaining Features: 15	AUC=0.9973	ACC=0.9835
Remaining Features: 14	AUC=0.9971	ACC=0.9817
Remaining Features: 13	AUC=0.9966	ACC=0.9823
Remaining Features: 12	AUC=0.9960	ACC=0.9737
Remaining Features: 11	AUC=0.9962	ACC=0.9762
Remaining Features: 10	AUC=0.9914	ACC=0.9554
Remaining Features: 9	AUC=0.9904	ACC=0.9645
Remaining Features: 8	AUC=0.9910	ACC=0.9535
Remaining Features: 7	AUC=0.9889	ACC=0.9535
Remaining Features: 6	AUC=0.9858	ACC=0.9487
Remaining Features: 5	AUC=0.9836	ACC=0.9535
Remaining Features: 4	AUC=0.9598	ACC=0.9144
Remaining Features: 3	AUC=0.8416	ACC=0.7958
Remaining Features: 2	AUC=0.8189	ACC=0.7891

XGB Best Subset

Features : 20
Best AUC : 0.9976
Best ACC : 0.9859

Evaluating learner: RF

Remaining Features: 53	AUC=0.9960	ACC=0.9682
Remaining Features: 52	AUC=0.9960	ACC=0.9700
Remaining Features: 51	AUC=0.9962	ACC=0.9707
Remaining Features: 50	AUC=0.9963	ACC=0.9682
Remaining Features: 49	AUC=0.9961	ACC=0.9694
Remaining Features: 48	AUC=0.9960	ACC=0.9737
Remaining Features: 47	AUC=0.9957	ACC=0.9664
Remaining Features: 46	AUC=0.9960	ACC=0.9707
Remaining Features: 45	AUC=0.9957	ACC=0.9688
Remaining Features: 44	AUC=0.9965	ACC=0.9725
Remaining Features: 43	AUC=0.9959	ACC=0.9682
Remaining Features: 42	AUC=0.9960	ACC=0.9707
Remaining Features: 41	AUC=0.9958	ACC=0.9676
Remaining Features: 40	AUC=0.9959	ACC=0.9768
Remaining Features: 39	AUC=0.9964	ACC=0.9737
Remaining Features: 38	AUC=0.9962	ACC=0.9713
Remaining Features: 37	AUC=0.9967	ACC=0.9755
Remaining Features: 36	AUC=0.9960	ACC=0.9731
Remaining Features: 35	AUC=0.9961	ACC=0.9713
Remaining Features: 34	AUC=0.9960	ACC=0.9707
Remaining Features: 33	AUC=0.9964	ACC=0.9743
Remaining Features: 32	AUC=0.9963	ACC=0.9731
Remaining Features: 31	AUC=0.9961	ACC=0.9743
Remaining Features: 30	AUC=0.9962	ACC=0.9713
Remaining Features: 29	AUC=0.9953	ACC=0.9719
Remaining Features: 28	AUC=0.9961	ACC=0.9725
Remaining Features: 27	AUC=0.9954	ACC=0.9694
Remaining Features: 26	AUC=0.9958	ACC=0.9688
Remaining Features: 25	AUC=0.9954	ACC=0.9707
Remaining Features: 24	AUC=0.9955	ACC=0.9700
Remaining Features: 23	AUC=0.9955	ACC=0.9707
Remaining Features: 22	AUC=0.9958	ACC=0.9713
Remaining Features: 21	AUC=0.9960	ACC=0.9694
Remaining Features: 20	AUC=0.9956	ACC=0.9670
Remaining Features: 19	AUC=0.9954	ACC=0.9670
Remaining Features: 18	AUC=0.9953	ACC=0.9664
Remaining Features: 17	AUC=0.9953	ACC=0.9688
Remaining Features: 16	AUC=0.9950	ACC=0.9682
Remaining Features: 15	AUC=0.9952	ACC=0.9682
Remaining Features: 14	AUC=0.9952	ACC=0.9719
Remaining Features: 13	AUC=0.9953	ACC=0.9700
Remaining Features: 12	AUC=0.9940	ACC=0.9664

Remaining Features: 11 | AUC=0.9926 | ACC=0.9621
Remaining Features: 10 | AUC=0.9930 | ACC=0.9603
Remaining Features: 9 | AUC=0.9913 | ACC=0.9572
Remaining Features: 8 | AUC=0.9896 | ACC=0.9517
Remaining Features: 7 | AUC=0.9910 | ACC=0.9548
Remaining Features: 6 | AUC=0.9831 | ACC=0.9395
Remaining Features: 5 | AUC=0.9833 | ACC=0.9370
Remaining Features: 4 | AUC=0.9828 | ACC=0.9352
Remaining Features: 3 | AUC=0.9762 | ACC=0.9199
Remaining Features: 2 | AUC=0.9686 | ACC=0.9114

RF Best Subset

Features : 41
Best AUC : 0.9959
Best ACC : 0.9768

=====

bRFE Completed

=====

Selected Learner : XGB
Selected Features: 20
Best AUC : 0.9976
Best Accuracy : 0.9859

=====

Running ESOM...

=====

Efficient SMOTE-based Oversampling Method (ESOM)

=====

Original Class Distribution
Class 0: 906
Class 1: 730

Majority Class : 0
Minority Class : 1
Class Imbalance: 19.43%

Applying Adaptive SMOTE
k_neighbors = 5

Balanced Class Distribution
Class 0: 906
Class 1: 906

Generated Samples : 176

=====

Running EOLF...

=====

Evaluating : RF
Best AUC : 0.9966
Best ACC : 0.9774

=====

Evaluating : XGB
Best AUC : 0.9979
Best ACC : 0.9834

=====

Evaluating : LightGBM

Best AUC : 0.9986
Best ACC : 0.9834

=====

EOLF Completed

=====

	Model	AUC	Accuracy \
0	RF	0.996626	0.977369
1	XGB	0.997922	0.983439
2	LightGBM	0.998568	0.983441

BestParameters

0	{'max_depth': 20, 'min_samples_leaf': 1, 'min_...
1	{'learning_rate': 0.1, 'max_depth': 3, 'n_esti...
2	{'learning_rate': 0.1, 'n_estimators': 200, 'n...

Selected Model : LightGBM
Best AUC : 0.9986
Best ACC : 0.9834

Training Final Model...

Fold Results

Model	: LightGBM
Features	: 20
Accuracy	: 0.9780
AUC	: 0.9993
Precision	: 0.9875
Recall	: 0.9634
Specificity	: 0.9900
F1-score	: 0.9753
PR-AUC	: 0.9991

=====

Outer Fold 9

=====

Running bRFE...

=====

Best Recursive Feature Elimination (bRFE)

=====

Evaluating learner: XGB

Remaining Features: 53	AUC=0.9975	ACC=0.9823
Remaining Features: 52	AUC=0.9974	ACC=0.9841
Remaining Features: 51	AUC=0.9975	ACC=0.9811
Remaining Features: 50	AUC=0.9976	ACC=0.9798
Remaining Features: 49	AUC=0.9977	ACC=0.9811
Remaining Features: 48	AUC=0.9976	ACC=0.9817
Remaining Features: 47	AUC=0.9979	ACC=0.9841
Remaining Features: 46	AUC=0.9975	ACC=0.9835
Remaining Features: 45	AUC=0.9975	ACC=0.9811
Remaining Features: 44	AUC=0.9978	ACC=0.9829
Remaining Features: 43	AUC=0.9976	ACC=0.9811
Remaining Features: 42	AUC=0.9980	ACC=0.9829
Remaining Features: 41	AUC=0.9979	ACC=0.9811
Remaining Features: 40	AUC=0.9975	ACC=0.9829
Remaining Features: 39	AUC=0.9977	ACC=0.9823
Remaining Features: 38	AUC=0.9976	ACC=0.9798
Remaining Features: 37	AUC=0.9972	ACC=0.9829
Remaining Features: 36	AUC=0.9974	ACC=0.9804
Remaining Features: 35	AUC=0.9976	ACC=0.9841
Remaining Features: 34	AUC=0.9976	ACC=0.9829
Remaining Features: 33	AUC=0.9975	ACC=0.9841

Remaining Features: 32	AUC=0.9979	ACC=0.9823
Remaining Features: 31	AUC=0.9982	ACC=0.9829
Remaining Features: 30	AUC=0.9978	ACC=0.9817
Remaining Features: 29	AUC=0.9979	ACC=0.9823
Remaining Features: 28	AUC=0.9979	ACC=0.9835
Remaining Features: 27	AUC=0.9975	ACC=0.9841
Remaining Features: 26	AUC=0.9978	ACC=0.9823
Remaining Features: 25	AUC=0.9975	ACC=0.9829
Remaining Features: 24	AUC=0.9977	ACC=0.9823
Remaining Features: 23	AUC=0.9976	ACC=0.9841
Remaining Features: 22	AUC=0.9975	ACC=0.9811
Remaining Features: 21	AUC=0.9978	ACC=0.9804
Remaining Features: 20	AUC=0.9979	ACC=0.9811
Remaining Features: 19	AUC=0.9980	ACC=0.9798
Remaining Features: 18	AUC=0.9982	ACC=0.9829
Remaining Features: 17	AUC=0.9978	ACC=0.9805
Remaining Features: 16	AUC=0.9977	ACC=0.9804
Remaining Features: 15	AUC=0.9977	ACC=0.9829
Remaining Features: 14	AUC=0.9973	ACC=0.9829
Remaining Features: 13	AUC=0.9969	ACC=0.9805
Remaining Features: 12	AUC=0.9969	ACC=0.9798
Remaining Features: 11	AUC=0.9969	ACC=0.9829
Remaining Features: 10	AUC=0.9968	ACC=0.9786
Remaining Features: 9	AUC=0.9969	ACC=0.9792
Remaining Features: 8	AUC=0.9966	ACC=0.9780
Remaining Features: 7	AUC=0.9960	ACC=0.9756
Remaining Features: 6	AUC=0.9956	ACC=0.9707
Remaining Features: 5	AUC=0.9955	ACC=0.9707
Remaining Features: 4	AUC=0.9841	ACC=0.9499
Remaining Features: 3	AUC=0.9594	ACC=0.9120
Remaining Features: 2	AUC=0.8240	ACC=0.7941

XGB Best Subset

Features : 24
Best AUC : 0.9976
Best ACC : 0.9841

Evaluating learner: RF

Remaining Features: 53	AUC=0.9965	ACC=0.9725
Remaining Features: 52	AUC=0.9968	ACC=0.9750
Remaining Features: 51	AUC=0.9968	ACC=0.9749
Remaining Features: 50	AUC=0.9971	ACC=0.9743
Remaining Features: 49	AUC=0.9963	ACC=0.9725
Remaining Features: 48	AUC=0.9962	ACC=0.9713
Remaining Features: 47	AUC=0.9967	ACC=0.9731
Remaining Features: 46	AUC=0.9967	ACC=0.9750
Remaining Features: 45	AUC=0.9964	ACC=0.9707
Remaining Features: 44	AUC=0.9965	ACC=0.9719
Remaining Features: 43	AUC=0.9965	ACC=0.9743
Remaining Features: 42	AUC=0.9966	ACC=0.9756
Remaining Features: 41	AUC=0.9968	ACC=0.9768
Remaining Features: 40	AUC=0.9972	ACC=0.9749
Remaining Features: 39	AUC=0.9970	ACC=0.9731
Remaining Features: 38	AUC=0.9968	ACC=0.9750
Remaining Features: 37	AUC=0.9969	ACC=0.9719
Remaining Features: 36	AUC=0.9967	ACC=0.9731
Remaining Features: 35	AUC=0.9970	ACC=0.9725
Remaining Features: 34	AUC=0.9974	ACC=0.9762
Remaining Features: 33	AUC=0.9973	ACC=0.9749
Remaining Features: 32	AUC=0.9970	ACC=0.9774
Remaining Features: 31	AUC=0.9972	ACC=0.9792
Remaining Features: 30	AUC=0.9971	ACC=0.9768
Remaining Features: 29	AUC=0.9963	ACC=0.9707
Remaining Features: 28	AUC=0.9967	ACC=0.9762

Remaining Features: 27 | AUC=0.9967 | ACC=0.9756
Remaining Features: 26 | AUC=0.9961 | ACC=0.9725
Remaining Features: 25 | AUC=0.9964 | ACC=0.9719
Remaining Features: 24 | AUC=0.9960 | ACC=0.9749
Remaining Features: 23 | AUC=0.9964 | ACC=0.9725
Remaining Features: 22 | AUC=0.9962 | ACC=0.9749
Remaining Features: 21 | AUC=0.9966 | ACC=0.9756
Remaining Features: 20 | AUC=0.9964 | ACC=0.9725
Remaining Features: 19 | AUC=0.9955 | ACC=0.9743
Remaining Features: 18 | AUC=0.9959 | ACC=0.9725
Remaining Features: 17 | AUC=0.9961 | ACC=0.9731
Remaining Features: 16 | AUC=0.9958 | ACC=0.9731
Remaining Features: 15 | AUC=0.9956 | ACC=0.9750
Remaining Features: 14 | AUC=0.9958 | ACC=0.9756
Remaining Features: 13 | AUC=0.9955 | ACC=0.9743
Remaining Features: 12 | AUC=0.9939 | ACC=0.9694
Remaining Features: 11 | AUC=0.9941 | ACC=0.9646
Remaining Features: 10 | AUC=0.9936 | ACC=0.9615
Remaining Features: 9 | AUC=0.9914 | ACC=0.9572
Remaining Features: 8 | AUC=0.9918 | ACC=0.9603
Remaining Features: 7 | AUC=0.9902 | ACC=0.9572
Remaining Features: 6 | AUC=0.9866 | ACC=0.9450
Remaining Features: 5 | AUC=0.9853 | ACC=0.9456
Remaining Features: 4 | AUC=0.9845 | ACC=0.9371
Remaining Features: 3 | AUC=0.9808 | ACC=0.9377
Remaining Features: 2 | AUC=0.9675 | ACC=0.9108

RF Best Subset

Features : 32
Best AUC : 0.9972
Best ACC : 0.9792

=====

bRFE Completed

=====

Selected Learner : XGB
Selected Features: 24
Best AUC : 0.9976
Best Accuracy : 0.9841

=====

Running ESOM...

=====

Efficient SMOTE-based Oversampling Method (ESOM)

=====

Original Class Distribution
Class 0: 906
Class 1: 731

Majority Class : 0
Minority Class : 1
Class Imbalance: 19.32%

Applying Adaptive SMOTE
k_neighbors = 5

Balanced Class Distribution
Class 0: 906
Class 1: 906

Generated Samples : 175

Running EOLF...

Evaluating : RF
Best AUC : 0.9976
Best ACC : 0.9763

Evaluating : XGB
Best AUC : 0.9985
Best ACC : 0.984

Evaluating : LightGBM
Best AUC : 0.9984
Best ACC : 0.9884

EOLF Completed

	Model	AUC	Accuracy \
0	RF	0.997626	0.976269
1	XGB	0.998485	0.983996
2	LightGBM	0.998381	0.988418

BestParameters

0	{'max_depth': None, 'min_samples_leaf': 1, 'mi...
1	{'learning_rate': 0.1, 'max_depth': 3, 'n_esti...
2	{'learning_rate': 0.05, 'n_estimators': 200, '...

Selected Model : XGB
Best AUC : 0.9985
Best ACC : 0.984

Training Final Model...

Fold Results

Model	: XGB
Features	: 24
Accuracy	: 0.9834
AUC	: 0.9996
Precision	: 0.9875
Recall	: 0.9753
Specificity	: 0.9900
F1-score	: 0.9814
PR-AUC	: 0.9995

Outer Fold 10

Running bRFE...

Best Recursive Feature Elimination (bRFE)

Evaluating learner: XGB

Remaining Features: 53	AUC=0.9977	ACC=0.9853
Remaining Features: 52	AUC=0.9972	ACC=0.9835
Remaining Features: 51	AUC=0.9975	ACC=0.9829
Remaining Features: 50	AUC=0.9975	ACC=0.9872
Remaining Features: 49	AUC=0.9976	ACC=0.9866

Remaining Features: 48	AUC=0.9973	ACC=0.9853
Remaining Features: 47	AUC=0.9974	ACC=0.9847
Remaining Features: 46	AUC=0.9978	ACC=0.9829
Remaining Features: 45	AUC=0.9977	ACC=0.9853
Remaining Features: 44	AUC=0.9978	ACC=0.9847
Remaining Features: 43	AUC=0.9978	ACC=0.9829
Remaining Features: 42	AUC=0.9974	ACC=0.9835
Remaining Features: 41	AUC=0.9976	ACC=0.9860
Remaining Features: 40	AUC=0.9977	ACC=0.9878
Remaining Features: 39	AUC=0.9976	ACC=0.9860
Remaining Features: 38	AUC=0.9977	ACC=0.9841
Remaining Features: 37	AUC=0.9979	ACC=0.9847
Remaining Features: 36	AUC=0.9976	ACC=0.9853
Remaining Features: 35	AUC=0.9977	ACC=0.9835
Remaining Features: 34	AUC=0.9978	ACC=0.9841
Remaining Features: 33	AUC=0.9973	ACC=0.9817
Remaining Features: 32	AUC=0.9976	ACC=0.9847
Remaining Features: 31	AUC=0.9976	ACC=0.9872
Remaining Features: 30	AUC=0.9977	ACC=0.9841
Remaining Features: 29	AUC=0.9978	ACC=0.9860
Remaining Features: 28	AUC=0.9981	ACC=0.9823
Remaining Features: 27	AUC=0.9972	ACC=0.9841
Remaining Features: 26	AUC=0.9976	ACC=0.9835
Remaining Features: 25	AUC=0.9980	ACC=0.9853
Remaining Features: 24	AUC=0.9977	ACC=0.9841
Remaining Features: 23	AUC=0.9976	ACC=0.9853
Remaining Features: 22	AUC=0.9976	ACC=0.9823
Remaining Features: 21	AUC=0.9977	ACC=0.9872
Remaining Features: 20	AUC=0.9974	ACC=0.9853
Remaining Features: 19	AUC=0.9979	ACC=0.9860
Remaining Features: 18	AUC=0.9977	ACC=0.9878
Remaining Features: 17	AUC=0.9981	ACC=0.9866
Remaining Features: 16	AUC=0.9980	ACC=0.9866
Remaining Features: 15	AUC=0.9983	ACC=0.9866
Remaining Features: 14	AUC=0.9978	ACC=0.9823
Remaining Features: 13	AUC=0.9980	ACC=0.9823
Remaining Features: 12	AUC=0.9974	ACC=0.9817
Remaining Features: 11	AUC=0.9974	ACC=0.9811
Remaining Features: 10	AUC=0.9975	ACC=0.9780
Remaining Features: 9	AUC=0.9972	ACC=0.9805
Remaining Features: 8	AUC=0.9979	ACC=0.9823
Remaining Features: 7	AUC=0.9978	ACC=0.9829
Remaining Features: 6	AUC=0.9948	ACC=0.9743
Remaining Features: 5	AUC=0.9956	ACC=0.9689
Remaining Features: 4	AUC=0.9917	ACC=0.9597
Remaining Features: 3	AUC=0.9751	ACC=0.9377
Remaining Features: 2	AUC=0.9419	ACC=0.8833

XGB Best Subset

Features : 19
Best AUC : 0.9977
Best ACC : 0.9878

Evaluating learner: RF

Remaining Features: 53	AUC=0.9961	ACC=0.9707
Remaining Features: 52	AUC=0.9958	ACC=0.9682
Remaining Features: 51	AUC=0.9963	ACC=0.9713
Remaining Features: 50	AUC=0.9958	ACC=0.9756
Remaining Features: 49	AUC=0.9957	ACC=0.9725
Remaining Features: 48	AUC=0.9960	ACC=0.9707
Remaining Features: 47	AUC=0.9960	ACC=0.9707
Remaining Features: 46	AUC=0.9961	ACC=0.9731
Remaining Features: 45	AUC=0.9956	ACC=0.9713
Remaining Features: 44	AUC=0.9959	ACC=0.9737

Remaining Features: 43 | AUC=0.9962 | ACC=0.9750
Remaining Features: 42 | AUC=0.9954 | ACC=0.9688
Remaining Features: 41 | AUC=0.9964 | ACC=0.9756
Remaining Features: 40 | AUC=0.9960 | ACC=0.9731
Remaining Features: 39 | AUC=0.9961 | ACC=0.9725
Remaining Features: 38 | AUC=0.9959 | ACC=0.9707
Remaining Features: 37 | AUC=0.9965 | ACC=0.9756
Remaining Features: 36 | AUC=0.9962 | ACC=0.9731
Remaining Features: 35 | AUC=0.9962 | ACC=0.9743
Remaining Features: 34 | AUC=0.9967 | ACC=0.9743
Remaining Features: 33 | AUC=0.9964 | ACC=0.9756
Remaining Features: 32 | AUC=0.9963 | ACC=0.9719
Remaining Features: 31 | AUC=0.9965 | ACC=0.9762
Remaining Features: 30 | AUC=0.9968 | ACC=0.9750
Remaining Features: 29 | AUC=0.9962 | ACC=0.9731
Remaining Features: 28 | AUC=0.9954 | ACC=0.9707
Remaining Features: 27 | AUC=0.9960 | ACC=0.9719
Remaining Features: 26 | AUC=0.9958 | ACC=0.9713
Remaining Features: 25 | AUC=0.9955 | ACC=0.9676
Remaining Features: 24 | AUC=0.9951 | ACC=0.9707
Remaining Features: 23 | AUC=0.9959 | ACC=0.9731
Remaining Features: 22 | AUC=0.9962 | ACC=0.9756
Remaining Features: 21 | AUC=0.9958 | ACC=0.9725
Remaining Features: 20 | AUC=0.9959 | ACC=0.9743
Remaining Features: 19 | AUC=0.9946 | ACC=0.9695
Remaining Features: 18 | AUC=0.9954 | ACC=0.9737
Remaining Features: 17 | AUC=0.9956 | ACC=0.9725
Remaining Features: 16 | AUC=0.9954 | ACC=0.9719
Remaining Features: 15 | AUC=0.9952 | ACC=0.9689
Remaining Features: 14 | AUC=0.9955 | ACC=0.9750
Remaining Features: 13 | AUC=0.9944 | ACC=0.9713
Remaining Features: 12 | AUC=0.9941 | ACC=0.9658
Remaining Features: 11 | AUC=0.9936 | ACC=0.9621
Remaining Features: 10 | AUC=0.9938 | ACC=0.9627
Remaining Features: 9 | AUC=0.9930 | ACC=0.9560
Remaining Features: 8 | AUC=0.9932 | ACC=0.9609
Remaining Features: 7 | AUC=0.9901 | ACC=0.9572
Remaining Features: 6 | AUC=0.9893 | ACC=0.9566
Remaining Features: 5 | AUC=0.9834 | ACC=0.9438
Remaining Features: 4 | AUC=0.9844 | ACC=0.9450
Remaining Features: 3 | AUC=0.9812 | ACC=0.9334
Remaining Features: 2 | AUC=0.9877 | ACC=0.9548

RF Best Subset

Features : 15
Best AUC : 0.9955
Best ACC : 0.975

=====
bRFE Completed

=====
Selected Learner : XGB
Selected Features: 19
Best AUC : 0.9977
Best Accuracy : 0.9878
=====

Running ESOM...

=====
Efficient SMOTE-based Oversampling Method (ESOM)
=====

Original Class Distribution

Class 0: 906

Class 1: 731

Majority Class : 0

Minority Class : 1

Class Imbalance: 19.32%

Applying Adaptive SMOTE

k_neighbors = 5

Balanced Class Distribution

Class 0: 906

Class 1: 906

Generated Samples : 175

Running EOLF...

Evaluating : RF

Best AUC : 0.9977

Best ACC : 0.9807

Evaluating : XGB

Best AUC : 0.9986

Best ACC : 0.9873

Evaluating : LightGBM

Best AUC : 0.999

Best ACC : 0.9912

EOLF Completed

	Model	AUC	Accuracy \
0	RF	0.997705	0.980678
1	XGB	0.998555	0.987303
2	LightGBM	0.999007	0.991168

BestParameters

0 {'max_depth': None, 'min_samples_leaf': 1, 'mi...

1 {'learning_rate': 0.1, 'max_depth': 3, 'n_esti...

2 {'learning_rate': 0.1, 'n_estimators': 200, 'n...

Selected Model : LightGBM

Best AUC : 0.999

Best ACC : 0.9912

Training Final Model...

Fold Results

Model : LightGBM

Features : 19

Accuracy : 0.9945

AUC : 1.0000

Precision : 0.9878

Recall : 1.0000

Specificity : 0.9900

F1-score : 0.9939

PR-AUC : 1.0000

=====
===
Overall Performance
=====

=====
Fold Model Features Accuracy AUC Precision Recall \
0 1 LightGBM 17 0.978022 0.999389 0.975309 0.975309
1 2 LightGBM 15 0.972527 0.991199 0.952381 0.987654
2 3 LightGBM 19 0.989011 0.999756 1.000000 0.975309
3 4 XGB 20 0.989011 0.998900 0.987654 0.987654
4 5 LightGBM 16 0.978022 0.996577 0.975309 0.975309
5 6 LightGBM 19 0.983516 0.999633 0.987500 0.975309
6 7 LightGBM 17 0.983516 0.999512 0.987654 0.975610
7 8 LightGBM 20 0.978022 0.999268 0.987500 0.963415
8 9 XGB 24 0.983425 0.999630 0.987500 0.975309
9 10 LightGBM 19 0.994475 1.000000 0.987805 1.000000

Specificity F1 PR_AUC
0 0.980198 0.975309 0.999247
1 0.960396 0.969697 0.989141
2 1.000000 0.987500 0.999703
3 0.990099 0.987654 0.998738
4 0.980198 0.975309 0.995097
5 0.990099 0.981366 0.999543
6 0.990000 0.981595 0.999416
7 0.990000 0.975309 0.999118
8 0.990000 0.981366 0.999543
9 0.990000 0.993865 1.000000

Average Performance
Fold 5.500000
Features 18.600000
Accuracy 0.982955
AUC 0.998386
Precision 0.982861
Recall 0.979088
Specificity 0.986099
F1 0.980897
PR_AUC 0.997954
dtype: float64

Standard Deviation
Fold 3.027650
Features 2.547330
Accuracy 0.006571
AUC 0.002704
Precision 0.012788
Recall 0.010073
Specificity 0.010627
F1 0.007284
PR_AUC 0.003398
dtype: float64